

Chapter 8

Large Deviations for Sparse Graphs

The development in this monograph till now has been based on results from graph limit theory. This theory, however, is inadequate for understanding the behavior of sparse graphs. The goal of this chapter is to describe an alternative approach, called nonlinear large deviations, that allows us to prove similar results for sparse graphs. Nonlinear large deviation theory gives a way of getting quantitative error bounds in some of the large deviation theorems proved in earlier chapters. The quantitative error bounds make it possible to extend the results to the sparse regime. At the time of writing this monograph, this theory is not as well-developed as the theory for dense graphs, but developed enough to make some progress about sparse graphs.

8.1 An Approximation for Partition Functions

Take any $N \geq 1$. Let $\|f\|$ denote the supremum norm of any function $f : [0, 1]^N \rightarrow \mathbb{R}$. Suppose that $f : [0, 1]^N \rightarrow \mathbb{R}$ is twice continuously differentiable in $(0, 1)^N$, such that f and all its first and second order derivatives extend continuously to the boundary. For each i and j , let

$$f_i := \frac{\partial f}{\partial x_i} \quad \text{and} \quad f_{ij} := \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

Define

$$a := \|f\|, \quad b_i := \|f_i\| \quad \text{and} \quad c_{ij} := \|f_{ij}\|.$$

Given $\epsilon > 0$, let $\mathcal{D}(\epsilon)$ be a finite subset of $\mathbb{R}^N$ such that for all $x \in \{0, 1\}^N$, there exists $d = (d_1, \dots, d_N) \in \mathcal{D}(\epsilon)$ such that

$$\sum_{i=1}^N (f_i(x) - d_i)^2 \leq N\epsilon^2. \tag{8.1.1}$$

Define

$$F := \log \sum_{x \in \{0,1\}^N} e^{f(x)}.$$

In the terminology of statistical mechanics, F is the logarithm of the partition function of the probability measure on $\{0, 1\}^N$ with Hamiltonian f . For $x \in [0, 1]$, let

$$I(x) := x \log x + (1 - x) \log(1 - x), \quad (8.1.2)$$

and for $x = (x_1, \dots, x_N) \in [0, 1]^N$, let

$$I(x) := \sum_{i=1}^N I(x_i).$$

The following theorem gives a sufficient condition on f under which the approximation

$$F = \sup_{x \in [0,1]^N} (f(x) - I(x)) + \text{lower order terms} \quad (8.1.3)$$

is valid. This is sometimes called the ‘naive mean field approximation’ in the statistical physics literature. Roughly speaking, the condition is that in addition to some smoothness assumptions, the gradient vector $\nabla f(x) = (\partial f / \partial x_1, \dots, \partial f / \partial x_N)$ may be approximately encoded by $o(N)$ bits of information. A bit more precisely, we need $|\mathcal{D}(\epsilon)| = e^{o(N)}$ for some $\epsilon = o(1)$, where the implicit assumption is that $N \rightarrow \infty$ and f varies with N . We will refer to this as the ‘low complexity gradient’ condition.

Theorem 8.1 *Let f , a , b_i , c_{ij} , $\mathcal{D}(\epsilon)$, F and I be as above. Then for any $\epsilon > 0$,*

$$F \leq \sup_{x \in [0,1]^N} (f(x) - I(x)) + \text{complexity} + \text{smoothness},$$

where

$$\begin{aligned} \text{complexity} &= \frac{1}{4} \left(N \sum_{i=1}^N b_i^2 \right)^{1/2} \epsilon + 3N\epsilon + \log |\mathcal{D}(\epsilon)|, \text{ and} \\ \text{smoothness} &= 4 \left(\sum_{i=1}^N (ac_{ii} + b_i^2) + \frac{1}{4} \sum_{i,j=1}^N (ac_{ij}^2 + b_i b_j c_{ij} + 4b_i c_{ij}) \right)^{1/2} \\ &\quad + \frac{1}{4} \left(\sum_{i=1}^N b_i^2 \right)^{1/2} \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2} + 3 \sum_{i=1}^N c_{ii} + \log 2. \end{aligned}$$

Moreover, F satisfies the lower bound

$$F \geq \sup_{x \in [0,1]^N} (f(x) - I(x)) - \frac{1}{2} \sum_{i=1}^N c_{ii}.$$

The rest of this section is devoted to the proof of Theorem 8.1. Examples are worked in subsequent sections.

We will generally denote the i th coordinate of a vector $x \in \mathbb{R}^N$ by x_i . Similarly, the i th coordinate of a random vector X will be denoted by X_i . Given $x \in [0, 1]^N$, define $x^{(i)}$ to be the vector $(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_N)$. For a random vector X define $X^{(i)}$ similarly. Given a function $g : [0, 1]^N \rightarrow \mathbb{R}$, define the discrete derivative $\Delta_i g$ as

$$\begin{aligned} \Delta_i g(x) &:= g(x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_N) \\ &\quad - g(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_N). \end{aligned} \quad (8.1.4)$$

For each i , define a function $\hat{x}_i : [0, 1]^N \rightarrow [0, 1]$ as

$$\hat{x}_i(x) = \frac{1}{1 + e^{-\Delta_i f(x)}}.$$

Let $\hat{x} : [0, 1]^N \rightarrow [0, 1]^N$ be the vector-valued function whose i th coordinate function is $\hat{x}_i$. When the vector x is understood from the context, we will simply write $\hat{x}$ and $\hat{x}_i$ instead of $\hat{x}(x)$ and $\hat{x}_i(x)$. The proof of Theorem 8.1 requires two key lemmas.

Lemma 8.1 *Let $X = (X_1, \dots, X_N)$ be a random vector that has probability density proportional to $e^{f(x)}$ on $\{0, 1\}^N$ with respect to the counting measure. Let $\hat{X} = \hat{x}(X)$. Then*

$$\mathbb{E}[(f(X) - f(\hat{X}))^2] \leq \sum_{i=1}^N (ac_{ii} + b_i^2) + \frac{1}{4} \sum_{i,j=1}^N (ac_{ij}^2 + b_i b_j c_{ij}).$$

Proof It is easy to see that

$$\hat{x}_i(x) = \mathbb{E}(X_i \mid X_j = x_j, 1 \leq j \leq N, j \neq i).$$

Let $D := f(X) - f(\hat{X})$. Then clearly

$$|D| \leq 2a. \quad (8.1.5)$$

Define

$$h(x) := f(x) - f(\hat{x}(x)),$$

so that $D = h(X)$. Note that for $i \neq j$,

$$\frac{\partial \hat{x}_j}{\partial x_i} = \frac{e^{-\Delta_j f(x)}}{(1 + e^{-\Delta_j f(x)})^2} \int_0^1 f_{ij}(x_1, \dots, x_{j-1}, t, x_{j+1}, \dots, x_n) dt,$$

and for $i = j$, the above derivative is identically equal to zero. Since

$$\frac{e^{-x}}{(1 + e^{-x})^2} \leq \frac{1}{4}$$

for all $x \in \mathbb{R}$, this shows that for all i and j ,

$$\left\| \frac{\partial \hat{x}_j}{\partial x_i} \right\| \leq \frac{c_{ij}}{4}. \quad (8.1.6)$$

Thus,

$$\begin{aligned} \left\| \frac{\partial h}{\partial x_i} \right\| &\leq \|f_i\| + \sum_{j=1}^N \|f_j\| \left\| \frac{\partial \hat{x}_j}{\partial x_i} \right\| \\ &\leq b_i + \frac{1}{4} \sum_{j=1}^N b_j c_{ij}. \end{aligned} \quad (8.1.7)$$

Consequently, if $D_i := h(X^{(i)})$, then

$$|D - D_i| \leq b_i + \frac{1}{4} \sum_{j=1}^N b_j c_{ij}. \quad (8.1.8)$$

For $t \in [0, 1]$ and $x \in [0, 1]^N$ define

$$u_i(t, x) := f_i(tx + (1 - t)\hat{x}),$$

so that

$$h(x) = \int_0^1 \sum_{i=1}^N (x_i - \hat{x}_i) u_i(t, x) dt.$$

Thus,

$$\mathbb{E}(D^2) = \int_0^1 \sum_{i=1}^N \mathbb{E}((X_i - \hat{X}_i) u_i(t, X) D) dt. \quad (8.1.9)$$

Now,

$$\|u_i\| \leq b_i, \quad (8.1.10)$$

and by (8.1.6),

$$\begin{aligned} \left\| \frac{\partial u_i}{\partial x_i} \right\| &\leq t \|f_{ii}\| + (1-t) \sum_{j=1}^N \|f_{ij}\| \left\| \frac{\partial \hat{x}_j}{\partial x_i} \right\| \\ &\leq t c_{ii} + \frac{1-t}{4} \sum_{j=1}^N c_{ij}^2. \end{aligned} \quad (8.1.11)$$

The bounds (8.1.5), (8.1.8), (8.1.10) and (8.1.11) imply that

$$\begin{aligned} &|\mathbb{E}((X_i - \hat{X}_i)u_i(t, X)D) - \mathbb{E}((X_i - \hat{X}_i)u_i(t, X^{(i)})D_i)| \\ &\leq \mathbb{E}|(u_i(t, X) - u_i(t, X^{(i)}))D| + \mathbb{E}|u_i(t, X^{(i)})(D - D_i)| \\ &\leq 2atc_{ii} + \frac{a(1-t)}{2} \sum_{j=1}^N c_{ij}^2 + b_i^2 + \frac{1}{4} \sum_{j=1}^N b_i b_j c_{ij}. \end{aligned}$$

But $u_i(t, X^{(i)})D_i$ is a function of the random variables $(X_j)_{j \neq i}$ only. Therefore by the definition of $\hat{X}_i$,

$$\mathbb{E}((X_i - \hat{X}_i)u_i(t, X^{(i)})D_i) = 0.$$

Thus,

$$|\mathbb{E}((X_i - \hat{X}_i)u_i(t, X)D)| \leq 2atc_{ii} + \frac{a(1-t)}{2} \sum_{j=1}^N c_{ij}^2 + b_i^2 + \frac{1}{4} \sum_{j=1}^N b_i b_j c_{ij}.$$

Using this bound in (8.1.9) gives

$$\begin{aligned} \mathbb{E}(D^2) &\leq \int_0^1 \sum_{i=1}^N \left(2atc_{ii} + \frac{a(1-t)}{2} \sum_{j=1}^N c_{ij}^2 + b_i^2 + \frac{1}{4} \sum_{j=1}^N b_i b_j c_{ij} \right) dt \\ &= \sum_{i=1}^N (ac_{ii} + b_i^2) + \frac{1}{4} \sum_{i,j=1}^N (ac_{ij}^2 + b_i b_j c_{ij}), \end{aligned}$$

completing the proof. $\square$

Lemma 8.2 *Let all notation be as in Lemma 8.1. Then*

$$\mathbb{E} \left[\left(\sum_{i=1}^N (X_i - \hat{X}_i) \Delta_i f(X) \right)^2 \right] \leq \sum_{i=1}^N b_i^2 + \frac{1}{4} \sum_{i,j=1}^N b_i(b_j + 4)c_{ij}.$$

Proof Let g_i denote the function $\Delta_i f$, for notational simplicity. Note that

$$g_i(x) = \int_0^1 f_i(x_1, \dots, x_{i-1}, t, x_{i+1}, \dots, x_N) dt,$$

which shows that

$$\|g_i\| \leq \|f_i\| = b_i \quad (8.1.12)$$

and for all j ,

$$\left\| \frac{\partial g_i}{\partial x_j} \right\| \leq \|f_{ij}\| = c_{ij}. \quad (8.1.13)$$

Let

$$G(x) := \sum_{i=1}^N (x_i - \hat{x}_i(x)) g_i(x).$$

Then

$$\frac{\partial G}{\partial x_i} = \sum_{j=1}^N \left[\left(1_{\{j=i\}} - \frac{\partial \hat{x}_j}{\partial x_i} \right) g_j(x) + (x_j - \hat{x}_j) \frac{\partial g_j}{\partial x_i} \right]$$

and therefore by (8.1.6), (8.1.12) and (8.1.13),

$$\left\| \frac{\partial G}{\partial x_i} \right\| \leq b_i + \frac{1}{4} \sum_{j=1}^N c_{ij} b_j + \sum_{j=1}^N c_{ij}. \quad (8.1.14)$$

Note that for any x ,

$$|G(x) - G(x^{(i)})| \leq \left\| \frac{\partial G}{\partial x_i} \right\|. \quad (8.1.15)$$

Again, $g_i(X)$ and $G(X^{(i)})$ are both functions of $(X_j)_{j \neq i}$ only. Therefore

$$\mathbb{E}((X_i - \hat{X}_i) g_i(X) G(X^{(i)})) = 0. \quad (8.1.16)$$

Combining (8.1.14), (8.1.15) and (8.1.16) gives

$$\begin{aligned}\mathbb{E}(G(X)^2) &= \sum_{i=1}^N \mathbb{E}((X_i - \hat{X}_i)g_i(X)G(X)) \\ &\leq \sum_{i=1}^N b_i \left(b_i + \frac{1}{4} \sum_{j=1}^N c_{ij}b_j + \sum_{j=1}^N c_{ij} \right).\end{aligned}$$

This completes the proof of the lemma. $\square$

With the aid of Lemma 8.1 and 8.2, we are now ready to prove Theorem 8.1.

Proof (Proof of the Upper Bound in Theorem 8.1) For $x, y \in [0, 1]^N$, let

$$g(x, y) := \sum_{i=1}^N (x_i \log y_i + (1 - x_i) \log(1 - y_i)). \quad (8.1.17)$$

Note that

$$g(x, \hat{x}) - I(\hat{x}) = \sum_{i=1}^N (x_i - \hat{x}_i) \log \frac{\hat{x}_i}{1 - \hat{x}_i} = \sum_{i=1}^N (x_i - \hat{x}_i) \Delta_i f(x). \quad (8.1.18)$$

Let

$$B := 4 \left(\sum_{i=1}^N (ac_{ii} + b_i^2) + \frac{1}{4} \sum_{i,j=1}^N (ac_{ij}^2 + b_i b_j c_{ij} + 4b_i c_{ij}) \right)^{1/2}.$$

Let

$$A_1 := \{x \in \{0, 1\}^N : |I(\hat{x}) - g(x, \hat{x})| \leq B/2\},$$

and

$$A_2 := \{x \in \{0, 1\}^N : |f(x) - f(\hat{x})| \leq B/2\}.$$

Let $A = A_1 \cap A_2$. By Lemma 8.1 and the identity (8.1.18), $\mathbb{P}(X \notin A_1) \leq 1/4$. By Lemma 8.2, $\mathbb{P}(X \notin A_2) \leq 1/4$. Thus, $\mathbb{P}(X \in A) \geq 1/2$, which is the same as

$$\frac{\sum_{x \in A} e^{f(x)}}{\sum_{x \in \{0, 1\}^N} e^{f(x)}} \geq \frac{1}{2}.$$

Therefore by the definition of the set A ,

$$\begin{aligned} F &= \log \sum_{x \in \{0,1\}^N} e^{f(x)} \leq \log \sum_{x \in A} e^{f(x)} + \log 2 \\ &\leq B + \log \sum_{x \in A} e^{f(\hat{x}) - I(\hat{x}) + g(x, \hat{x})} + \log 2. \end{aligned} \quad (8.1.19)$$

Now take some $x \in \{0, 1\}^N$ and let d satisfy (8.1.1). Then by the Cauchy–Schwarz inequality,

$$\sum_{i=1}^N |f_i(x) - d_i| \leq N\epsilon.$$

Fix such an x and d . Note that for each i ,

$$\begin{aligned} |\Delta_i f(x) - f_i(x)| &\leq \int_0^1 |f_i(x_1, \dots, x_{i-1}, t, x_{i+1}, \dots, x_N) - f_i(x)| dt \\ &\leq \|f_{ii}\| = c_{ii}. \end{aligned}$$

By the last two inequalities and (8.1.1),

$$\sum_{i=1}^N |\Delta_i f(x) - d_i| \leq N\epsilon + \sum_{i=1}^N c_{ii}. \quad (8.1.20)$$

and

$$\left(\sum_{i=1}^N (\Delta_i f(x) - d_i)^2 \right)^{1/2} \leq N^{1/2}\epsilon + \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2}. \quad (8.1.21)$$

Let $u(x) = 1/(1 + e^{-x})$. Note that for all x ,

$$|u'(x)| = \frac{1}{(e^{x/2} + e^{-x/2})^2} \leq \frac{1}{4}.$$

Therefore if a vector $p = p(d)$ is defined as $p_i = u(d_i)$, then by (8.1.21),

$$\begin{aligned} \left(\sum_{i=1}^N (\hat{x}_i - p_i)^2 \right)^{1/2} &\leq \left(\frac{1}{16} \sum_{i=1}^N (\Delta_i f(x) - d_i)^2 \right)^{1/2} \\ &\leq \frac{N^{1/2}\epsilon}{4} + \frac{1}{4} \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2}. \end{aligned}$$

Thus, if

$$L := \left(\sum_{i=1}^N b_i^2 \right)^{1/2},$$

then

$$\begin{aligned} |f(\hat{x}) - f(p)| &\leq L \left(\sum_{i=1}^N (\hat{x}_i - p_i)^2 \right)^{1/2} \\ &\leq \frac{LN^{1/2}\epsilon}{4} + \frac{L}{4} \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2}. \end{aligned} \quad (8.1.22)$$

Next, let $v(x) = \log(1 + e^{-x})$. Then for all x ,

$$|v'(x)| = \frac{e^{-x}}{1 + e^{-x}} \leq 1.$$

Consequently,

$$|\log \hat{x}_i - \log p_i| \leq |\Delta_i f(x) - d_i|$$

and

$$|\log(1 - \hat{x}_i) - \log(1 - p_i)| \leq |\Delta_i f(x) - d_i|.$$

Therefore by (8.1.20),

$$|g(x, \hat{x}) - g(x, p)| \leq 2 \sum_{i=1}^N |\Delta_i f(x) - d_i| \leq 2N\epsilon + 2 \sum_{i=1}^N c_{ii}. \quad (8.1.23)$$

Finally, let $w(x) = I(u(x))$. Then

$$\begin{aligned} w'(x) &= u'(x)I'(u(x)) \\ &= \frac{e^{-x}}{(1 + e^{-x})^2} \log \frac{u(x)}{1 - u(x)} \\ &= \frac{xe^{-x}}{(1 + e^{-x})^2}. \end{aligned}$$

Thus, for all x ,

$$|w'(x)| \leq \sup_{x \in \mathbb{R}} \frac{|x|e^{-x}}{(1 + e^{-x})^2} \leq \sup_{x \geq 0} xe^{-x} = \frac{1}{e}.$$

Consequently,

$$|I(\hat{x}_i) - I(p_i)| \leq \frac{1}{e} |\Delta_i f(x) - d_i|,$$

and so by (8.1.20),

$$|I(\hat{x}) - I(p)| \leq \frac{N\epsilon}{e} + \frac{1}{e} \sum_{i=1}^N c_{ii}. \quad (8.1.24)$$

For each $d \in \mathcal{D}(\epsilon)$ let $\mathcal{C}(d)$ be the set of all $x \in \{0, 1\}^N$ such that (8.1.1) holds, and let $p(d)$ be the vector p defined above. Then by (8.1.22), (8.1.23) and (8.1.24),

$$\begin{aligned} \log \sum_{x \in A} e^{f(\hat{x}) - I(\hat{x}) + g(x, \hat{x})} &\leq \log \sum_{d \in \mathcal{D}(\epsilon)} \sum_{x \in \mathcal{C}(d)} e^{f(\hat{x}) - I(\hat{x}) + g(x, \hat{x})} \\ &\leq \frac{LN^{1/2}\epsilon}{4} + \frac{L}{4} \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2} + 2N\epsilon + 2 \sum_{i=1}^N c_{ii} + \frac{N\epsilon}{e} + \frac{1}{e} \sum_{i=1}^N c_{ii} \\ &\quad + \log \sum_{d \in \mathcal{D}(\epsilon)} \sum_{x \in \mathcal{C}(d)} e^{f(p(d)) - I(p(d)) + g(x, p(d))}. \end{aligned} \quad (8.1.25)$$

Now note that for any $p \in [0, 1]^N$,

$$\sum_{x \in \{0, 1\}^N} e^{g(x, p)} = 1.$$

Thus,

$$\begin{aligned} \log \sum_{d \in \mathcal{D}(\epsilon)} \sum_{x \in \mathcal{C}(d)} e^{f(p(d)) - I(p(d)) + g(x, p(d))} \\ \leq \log \sum_{d \in \mathcal{D}(\epsilon)} e^{f(p(d)) - I(p(d))} \\ \leq \log |\mathcal{D}(\epsilon)| + \sup_{p \in [0, 1]^N} (f(p) - I(p)). \end{aligned} \quad (8.1.26)$$

Combining (8.1.19), (8.1.25) and (8.1.26), the proof is complete. $\square$

Proof (Proof of the Lower Bound in Theorem 8.1) Let y be a point in the cube $[0, 1]^N$. Let $Y = (Y_1, \dots, Y_N)$ be a random vector with independent components, where Y_i is a Bernoulli(y_i) random variable. Then by Jensen's inequality,

$$\begin{aligned}
\sum_{x \in \{0,1\}^N} e^{f(x)} &= \sum_{x \in \{0,1\}^N} e^{f(x) - g(x,y) + g(x,y)} \\
&= \mathbb{E}(e^{f(Y) - g(Y,y)}) \\
&\geq \exp(\mathbb{E}(f(Y) - g(Y,y))) \\
&= \exp(\mathbb{E}(f(Y)) - I(y)).
\end{aligned}$$

Let $S := f(Y) - f(y)$. For $t \in [0, 1]$ and $x \in [0, 1]^N$ define

$$v_i(t, x) := f_i(tx + (1-t)y),$$

so that

$$S = \int_0^1 \sum_{i=1}^N (Y_i - y_i) v_i(t, Y) dt. \quad (8.1.27)$$

By the independence of Y_i and $Y^{(i)}$,

$$\begin{aligned}
|\mathbb{E}((Y_i - y_i) v_i(t, Y))| &= |\mathbb{E}((Y_i - y_i)(v_i(t, Y) - v_i(t, Y^{(i)})))| \\
&\leq \left\| \frac{\partial v_i}{\partial x_i} \right\| \leq t c_{ii}.
\end{aligned}$$

Using this bound in (8.1.27) gives

$$\mathbb{E}(S) \geq - \int_0^1 \sum_{i=1}^N t c_{ii} dt = - \frac{1}{2} \sum_{i=1}^N c_{ii}.$$

This completes the proof. $\square$

8.2 Gradient Complexity of Homomorphism Densities

The goal of this section is to compute estimates for the smoothness and complexity terms of Theorem 8.1 for homomorphism densities. These estimates will be used later for establishing quantitative versions of some of the theorems proved earlier in this monograph, which, in turn, will be used for understanding large deviations in sparse graphs.

Let $n \geq 2$ be a positive integer. Let $\mathcal{P}_n$ denote the set of all upper triangular arrays like $x = (x_{ij})_{1 \leq i < j \leq n}$, where each $x_{ij} \in [0, 1]$. Let H be a finite simple graph with vertex set $V = \{1, 2, \dots, k\}$ and edge set E . For $x \in \mathcal{P}_n$, let

$$t(H, x) := \frac{1}{n^k} \sum_{q_1, \dots, q_k=1}^n \prod_{\{l, l'\} \in E} x_{q_l q_{l'}}, \quad (8.2.1)$$

where x_{ii} is interpreted as zero for each i and x_{ij} denotes x_{ji} if $i > j$. Let

$$f(x) := n^2 t(H, x). \quad (8.2.2)$$

We will now obtain estimates for the smoothness and complexity terms for this f . The smoothness term is straightforward. The estimates for the smoothness term are given in Theorem 8.2 below. The bulk of this section is devoted to the analysis of the complexity term. The estimates for the complexity term are given in Theorem 8.3.

Theorem 8.2 *Let H and f be as above, and let $m := |E|$. Then $\|f\| \leq n^2$, and for any $i < j$ and $i' < j'$,*

$$\begin{aligned} \left\| \frac{\partial f}{\partial x_{ij}} \right\| &\leq 2m, \text{ and} \\ \left\| \frac{\partial^2 f}{\partial x_{ij} \partial x_{i'j'}} \right\| &\leq \begin{cases} 4m(m-1)n^{-1} & \text{if } |\{i, j, i', j'\}| = 2 \text{ or } 3, \\ 4m(m-1)n^{-2} & \text{if } |\{i, j, i', j'\}| = 4. \end{cases} \end{aligned}$$

Proof It is clear that $\|f\| \leq n^2$ since the x_{ij} 's are all in $[0, 1]$ and there are exactly n^k terms in the sum that defines f . Next, note that for any $i < j$,

$$\frac{\partial f}{\partial x_{ij}} = \frac{1}{n^{k-2}} \sum_{\{a,b\} \in E} \sum_{\substack{q \in [n]^k \\ \{q_a, q_b\} = \{i,j\}}} \prod_{\substack{\{l,l'\} \in E \\ \{l,l'\} \neq \{a,b\}}} x_{q_l q_{l'}}, \quad (8.2.3)$$

and therefore

$$\left\| \frac{\partial f}{\partial x_{ij}} \right\| \leq \frac{2mn^{k-2}}{n^{k-2}} = 2m.$$

Next, for any $i < j$ and $i' < j'$,

$$\frac{\partial^2 f}{\partial x_{ij} \partial x_{i'j'}} = \frac{1}{n^{k-2}} \sum_{\{a,b\} \in E} \sum_{\substack{\{c,d\} \in E \\ \{c,d\} \neq \{a,b\}}} \sum_{\substack{q \in [n]^k \\ \{q_a, q_b\} = \{i,j\} \\ \{q_c, q_d\} = \{i',j'\}}} \prod_{\substack{\{l,l'\} \in E \\ \{l,l'\} \neq \{a,b\} \\ \{l,l'\} \neq \{c,d\}}} x_{q_l q_{l'}}.$$

Take any two edges $\{a, b\}, \{c, d\} \in E$ such that $\{a, b\} \neq \{c, d\}$. Then the number of choices of $q \in [n]^k$ such that $\{q_a, q_b\} = \{i, j\}$ and $\{q_c, q_d\} = \{i', j'\}$ is at most $4n^{k-3}$ if $|\{i, j, i', j'\}| = 2$ or 3 (since we are constraining q_a, q_b, q_c and q_d and $|\{a, b, c, d\}| \geq 3$ always), and at most $4n^{k-4}$ if $|\{i, j, i', j'\}| = 4$ (since $|\{a, b, c, d\}|$ must be 4 if there is at least one possible choice of q for these i, j, i', j'). This gives the upper bound for the second derivatives. $\square$

To understand the complexity of the gradient of f , we need some preparation. For an $n \times n$ matrix M , recall the definition of the operator norm:

$$\|M\| := \max\{\|Mz\| : z \in \mathbb{R}^n, \|z\| = 1\},$$

where $\|z\|$ denotes the Euclidean norm of z . Equivalently, $\|M\|$ is the largest singular value of M . For $x \in \mathcal{P}_n$, let $M(x)$ be the symmetric matrix whose (i, j) th entry is x_{ij} , with the convention that $x_{ij} = x_{ji}$ and $x_{ii} = 0$. Define the operator norm of x as

$$\|x\|_{\text{op}} := \|M(x)\|.$$

The following lemma estimates the entropy of the unit cube under this norm.

Lemma 8.3 *For any $\tau \in (0, 1)$, there is a finite set of $n \times n$ matrices $\mathcal{W}(\tau)$ such that*

$$|\mathcal{W}(\tau)| \leq e^{34(n/\tau^2) \log(51/\tau^2)},$$

and for any $n \times n$ matrix M with entries in $[0, 1]$, there exists $W \in \mathcal{W}(\tau)$ such that

$$\|M - W\| \leq n\tau.$$

In particular, for any $x \in \mathcal{P}_n$ there exists $W \in \mathcal{W}(\tau)$ such that

$$\|M(x) - W\| \leq n\tau.$$

Proof Suppose that $n < 17/\tau^2$. Let $\mathcal{W}(\tau)$ consist of all $n \times n$ matrices whose entries are all integer multiples of τ and belong to the interval $[0, 1]$. It is then easy to see that for any M with entries in $[0, 1]$, there exists $W \in \mathcal{W}(\tau)$ such that $\|M - W\| \leq n\tau$. Moreover, since $n < 17/\tau^2$,

$$|\mathcal{W}(\tau)| \leq \tau^{-n^2} \leq e^{9(n/\tau^2) \log(1/\tau^2)},$$

completing the proof in this case.

Next, suppose that $n \geq 17/\tau^2$. Let l be the integer part of $17/\tau^2$ and $\delta = 1/l$. Let $\mathcal{A}$ be a finite subset of the unit ball of $\mathbb{R}^n$ such that any vector inside the ball is at Euclidean distance $\leq \delta$ from some element of $\mathcal{A}$. (In other words, $\mathcal{A}$ is a δ -net of the unit ball under the Euclidean metric.) The set $\mathcal{A}$ may be defined as a maximal set of points in the unit ball such that any two are at a distance greater than δ from each other. Since the balls of radius $\delta/2$ around these points are disjoint and their union is contained in the ball of radius $1 + \delta/2$ centered at zero, it follows that $|\mathcal{A}|C(\delta/2)^n \leq C(1 + \delta/2)^n$, where C is the volume of the unit ball. Therefore,

$$|\mathcal{A}| \leq (3/\delta)^n. \tag{8.2.4}$$

Take any $x \in \mathcal{P}_n$. Suppose that M has singular value decomposition

$$M = \sum_{i=1}^n \lambda_i u_i v_i^t,$$

where $\lambda_1 \geq \lambda_2 \geq \dots \lambda_n \geq 0$ are the singular values of M , and $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are singular vectors, and v_i^t denotes the transpose of the column vector v_i . Assume that the u_i 's and v_i 's are orthonormal systems. Since the elements of M all belong to the interval $[0, 1]$, it is easy to see that $\lambda_1 \leq n$ and $\sum \lambda_i^2 \leq n^2$. Due to the second inequality, there exists $y \in \mathcal{A}$ such that

$$\sum_{i=1}^n (n^{-1} \lambda_i - y_i)^2 \leq \delta^2. \quad (8.2.5)$$

Let $z_1, \dots, z_n$ and $w_1, \dots, w_n$ be elements of $\mathcal{A}$ such that for each i ,

$$\sum_{j=1}^n (u_{ij} - z_{ij})^2 \leq \delta^2 \quad \text{and} \quad \sum_{j=1}^n (v_{ij} - w_{ij})^2 \leq \delta^2, \quad (8.2.6)$$

where u_{ij} denotes the j th component of the vector u_i , etc. Define two matrices V and W as

$$V := \sum_{i=1}^{l-1} \lambda_i u_i v_i^t \quad \text{and} \quad W := \sum_{i=1}^{l-1} n y_i z_i w_i^t.$$

Note that since $\sum \lambda_i^2 \leq n^2$ and λ_i decreases with i , therefore for each i , $\lambda_i^2 \leq n^2/i$. Thus,

$$\begin{aligned} \|M - W\| &\leq \|M - V\| + \|V - W\| \\ &\leq \frac{n}{\sqrt{l}} + \|V - W\|. \end{aligned}$$

Next, note that by (8.2.6), the operator norms of the rank-one matrices $(u_i - z_i)v_i^t$ and $z_i(v_i - w_i)^t$ are bounded by δ . And by (8.2.5), $|\lambda_i - n y_i| \leq n\delta$ for each i . Therefore

$$\begin{aligned} \|V - W\| &\leq \left\| \sum_{i=1}^{l-1} (\lambda_i - n y_i) u_i v_i^t \right\| + \left\| \sum_{i=1}^{l-1} n y_i (u_i - z_i) v_i^t \right\| \\ &\quad + \left\| \sum_{i=1}^{l-1} n y_i z_i (v_i - w_i)^t \right\| \\ &\leq \max_{1 \leq i \leq l-1} |\lambda_i - n y_i| + 2 \sum_{i=1}^{l-1} n |y_i| \delta \\ &\leq n\delta + 2n\delta \left((l-1) \sum_{i=1}^{l-1} y_i^2 \right)^{1/2} \leq n\delta + 2n\delta \sqrt{l-1} \leq \frac{3n}{\sqrt{l}}. \end{aligned}$$

Thus,

$$\|M - W\| \leq \frac{4n}{\sqrt{l}} \leq \frac{4n}{\sqrt{\frac{17}{\tau^2} - 1}} \leq \frac{4n}{\sqrt{\frac{16}{\tau^2}}} = n\tau.$$

Let $\mathcal{W}(\tau)$ be the set of all possible W 's constructed in the above manner. Then $\mathcal{W}(\tau)$ has the required property, and by (8.2.4),

$$|\mathcal{W}(\tau)| \leq \text{The number of ways of choosing}$$

$$\begin{aligned} & y, z_1, \dots, z_{l-1}, w_1, \dots, w_{l-1} \in \mathcal{A} \\ &= |\mathcal{A}|^{2l-1} \leq (3/\delta)^{2nl} = e^{2nl \log(3l)}. \end{aligned}$$

This completes the proof of the lemma. $\square$

Let r be a positive integer. Let K_r be the complete graph on the vertex set $\{1, \dots, r\}$. For any set of edges A of K_r , any $q = (q_1, \dots, q_r) \in [n]^r$, and any $x \in \mathcal{P}_n$, let

$$P(x, q, A) := \prod_{\{a,b\} \in A} x_{q_a q_b},$$

with the usual convention that the empty product is 1. Note that if $q_a = q_b$ for some $\{a, b\} \in A$, the $P(x, q, A) = 0$ due to our convention that $x_{ii} = 0$ for each i . Next, note that if A and B are disjoint sets of edges, then

$$P(x, q, A \cup B) = P(x, q, A)P(x, q, B). \quad (8.2.7)$$

Lemma 8.4 *Let A and B be sets of edges of K_r , and let $e = \{\alpha, \beta\}$ be an edge that is neither in A nor in B . Then for any $x, y \in \mathcal{P}_n$,*

$$\left| \sum_{q \in [n]^r} P(x, q, A)P(y, q, B)(x_{q_\alpha q_\beta} - y_{q_\alpha q_\beta}) \right| \leq n^{r-1} \|x - y\|_{\text{op}}.$$

Proof By relabeling the vertices of K_r and redefining A and B , we may assume that $\alpha = 1$ and $\beta = 2$.

Let A_1 be the set of all edges in A that are incident to 1. Let A_2 be the set of all edges in A that are incident to 2. Note that since $\{1, 2\} \notin A$, therefore A_1 and A_2 must be disjoint. Similarly, let B_1 be the set of all edges in B that are incident to 1 and let B_2 be the set of all edges in B that are incident to 2. Let $A_3 = A \setminus (A_1 \cup A_2)$ and $B_3 = B \setminus (B_1 \cup B_2)$. By (8.2.7),

$$P(x, q, A) = P(x, q, A_1)P(x, q, A_2)P(x, q, A_3)$$

and

$$P(y, q, B) = P(y, q, B_1)P(y, q, B_2)P(y, q, B_3).$$

Thus,

$$\begin{aligned} & \sum_{q \in [n]^r} P(x, q, A)P(y, q, B)(x_{q_1 q_2} - y_{q_1 q_2}) \\ &= \sum_{q_3, \dots, q_r} P(x, q, A_3)P(y, q, B_3) \left(\sum_{q_1, q_2} Q(x, y, q)(x_{q_1 q_2} - y_{q_1 q_2}) \right), \end{aligned}$$

where

$$Q(x, y, q) = P(x, q, A_1)P(x, q, A_2)P(y, q, B_1)P(y, q, B_2).$$

Now fix $q_3, \dots, q_r$. Then $P(x, q, A_1)P(y, q, B_1)$ is a function of q_1 only, and does not depend on q_2 . Let $g(q_1)$ denote this function. Similarly, $P(x, q, A_2)P(y, q, B_2)$ is a function of q_2 only, and does not depend on q_1 . Let $h(q_2)$ denote this function. Both g and h are uniformly bounded by 1. Therefore

$$\begin{aligned} \left| \sum_{q_1, q_2} Q(x, y, q)(x_{q_1 q_2} - y_{q_1 q_2}) \right| &= \left| \sum_{q_1, q_2} g(q_1)h(q_2)(x_{q_1 q_2} - y_{q_1 q_2}) \right| \\ &\leq n \|x - y\|_{\text{op}}. \end{aligned}$$

Since this is true for all choices of $q_3, \dots, q_r$ and P is also uniformly bounded by 1, this completes the proof of the lemma. $\square$

Let A and B be two sets of edges of K_r . For $x, y \in \mathcal{P}_n$, define

$$R(x, y, A, B) := \sum_{q \in [n]^r} P(x, q, A)P(y, q, B).$$

Lemma 8.5 *Let A, B, A' and B' be sets of edges of K_r such that $A \cap B = A' \cap B' = \emptyset$ and $A \cup B = A' \cup B'$. Then*

$$|R(x, y, A, B) - R(x, y, A', B')| \leq \frac{1}{2} r(r-1) n^{r-1} \|x - y\|_{\text{op}}.$$

Proof First, suppose that $e = \{\alpha, \beta\}$ is an edge such that $e \notin A'$ and $A = A' \cup \{e\}$. Since $A \cup B = A' \cup B'$ and $A \cap B = A' \cap B' = \emptyset$, this implies that $e \notin B$ and $B' = B \cup \{e\}$. Thus,

$$R(x, y, A, B) - R(x, y, A', B') = \sum_{q \in [n]^r} P(x, q, A')P(y, q, B)(x_{q_\alpha q_\beta} - y_{q_\alpha q_\beta}),$$

and the proof is completed using Lemma 8.4. For the general case, simply ‘move’ from the pair (A, B) to the pair (A', B') by ‘moving one edge at a time’ and apply Lemma 8.4 at each step. $\square$

Lemma 8.6 *Let $g_{ij} := \partial f / \partial x_{ij}$, where f is the function defined in Eq. (8.2.2), and let $m := |E|$, as before. Then for any $x, y \in \mathcal{P}_n$,*

$$\sum_{1 \leq i < j \leq n} (g_{ij}(x) - g_{ij}(y))^2 \leq 8m^2 k^2 n \|x - y\|_{\text{op}}.$$

Proof Recall Eq. (8.2.3), that is, for any $1 \leq i < j \leq N$,

$$g_{ij}(x) = \frac{\partial f}{\partial x_{ij}} = \frac{1}{n^{k-2}} \sum_{\{a,b\} \in E} \sum_{\substack{q \in [n]^k \\ \{q_a, q_b\} = \{i,j\}}} \prod_{\substack{\{l,l'\} \in E \\ \{l,l'\} \neq \{a,b\}}} x_{q_l q_{l'}}.$$

Although differentiating with respect to x_{ii} does not make sense, let g_{ii} be the function defined using the same formula as above. When $i > j$, let $g_{ij} = g_{ji}$. Fix $x, y \in \mathcal{P}_n$. Define for any $q \in [n]^k$ and $\{a, b\} \in E$

$$D(q, \{a, b\}) := \prod_{\substack{\{l,l'\} \in E \\ \{l,l'\} \neq \{a,b\}}} x_{q_l q_{l'}} - \prod_{\substack{\{l,l'\} \in E \\ \{l,l'\} \neq \{a,b\}}} y_{q_l q_{l'}}.$$

Define

$$\theta_{ij} := \begin{cases} 2 & \text{if } i = j, \\ 1/2 & \text{if } i \neq j, \end{cases} \quad \gamma_{ij} := \begin{cases} 2 & \text{if } i = j, \\ 1 & \text{if } i \neq j. \end{cases}$$

Then note that

$$\begin{aligned} & \sum_{i,j=1}^n \theta_{ij} (g_{ij}(x) - g_{ij}(y))^2 \\ &= \frac{1}{n^{2k-4}} \sum_{i,j=1}^n \theta_{ij} \left(\sum_{\{a,b\} \in E} \sum_{\substack{q \in [n]^k \\ \{q_a, q_b\} = \{i,j\}}} D(q, \{a, b\}) \right)^2 \\ &= \frac{1}{n^{2k-4}} \sum_{i,j=1}^n \sum_{\substack{\{a,b\} \in E \\ \{c,d\} \in E}} \sum_{\substack{q \in [n]^k \\ \{q_a, q_b\} = \{i,j\}}} \sum_{\substack{s \in [n]^k \\ \{s_c, s_d\} = \{i,j\}}} \theta_{ij} D(q, \{a, b\}) D(s, \{c, d\}) \\ &= \frac{1}{n^{2k-4}} \sum_{\substack{\{a,b\} \in E \\ \{c,d\} \in E}} \sum_{q \in [n]^k} \sum_{\substack{s \in [n]^k \\ \{s_c, s_d\} = \{q_a, q_b\}}} \gamma_{q_a q_b} D(q, \{a, b\}) D(s, \{c, d\}). \end{aligned}$$

Now fix two edges $\{a, b\}$ and $\{c, d\}$ in E . Relabeling vertices if necessary, assume that $c = k - 1$ and $d = k$. Let $r = 2k - 2$. For any $t \in [n]^r$, define two vectors $q(t)$ and $s(t)$ in $[n]^k$ as follows. For $i = 1, \dots, k$, let $q_i(t) = t_i$. For $i = 1, \dots, k - 2$, let $s_i(t) = t_{i+k}$. Let $s_{k-1}(t) = t_a$ and $s_k(t) = t_b$. Note that

$$\begin{aligned} & \sum_{q \in [n]^k} \sum_{\substack{s \in [n]^k \\ \{s_c, s_d\} = \{q_a, q_b\}}} \gamma_{q_a q_b} D(q, \{a, b\}) D(s, \{c, d\}) \\ &= \sum_{q \in [n]^k} \sum_{\substack{s \in [n]^k \\ s_c = q_a, s_d = q_b}} D(q, \{a, b\}) D(s, \{c, d\}) \\ &+ \sum_{q \in [n]^k} \sum_{\substack{s \in [n]^k \\ s_c = q_b, s_d = q_a}} D(q, \{a, b\}) D(s, \{c, d\}). \end{aligned}$$

With the definition of $q(t)$ and $s(t)$ given above, it is clear that the first term on the right-hand side equals

$$\sum_{t \in [n]^r} D(q(t), \{a, b\}) D(s(t), \{c, d\}).$$

Below, we will get a bound on this term. The same upper bound will hold for the other term by symmetry.

Define two subsets of edges A and B of K_r as follows. Let A be the set of all edges $\{l, l'\}$ such that $\{l, l'\} \in E \setminus \{\{a, b\}\}$. Let B be the set of all edges $\{\phi(l), \phi(l')\}$ such that $\{l, l'\} \in E \setminus \{\{k-1, k\}\}$, where $\phi : [k] \rightarrow [r]$ is the map

$$\phi(x) = \begin{cases} x + k & \text{if } x \neq k-1 \text{ and } x \neq k, \\ a & \text{if } x = k-1, \\ b & \text{if } x = k. \end{cases}$$

By the above construction, $q_l(t) = t_l$ and $s_l(t) = t_{\phi(l)}$. Therefore it is easy to see, for instance, that

$$\begin{aligned} & \sum_{t \in [n]^r} \prod_{\substack{\{l, l'\} \in E \\ \{l, l'\} \neq \{a, b\}}} x_{q_l(t) q_{l'}(t)} \prod_{\substack{\{l, l'\} \in E \\ \{l, l'\} \neq \{k-1, k\}}} y_{s_l(t) s_{l'}(t)} \\ &= \sum_{t \in [n]^r} \prod_{\substack{\{l, l'\} \in E \\ \{l, l'\} \neq \{a, b\}}} x_{t_l t_{l'}} \prod_{\substack{\{l, l'\} \in E \\ \{l, l'\} \neq \{k-1, k\}}} y_{t_{\phi(l)} t_{\phi(l')}} \\ &= R(x, y, A, B). \end{aligned}$$

Carrying out similar computations for the other terms, we get

$$\begin{aligned} & \sum_{t \in [n]^r} D(q(t), \{a, b\}) D(s(t), \{c, d\}) \\ &= R(x, y, A \cup B, \emptyset) - R(x, y, B, A) - R(x, y, A, B) + R(x, y, \emptyset, A \cup B). \end{aligned}$$

Lastly, note that $A \cap B = \emptyset$ since for any $\{l, l'\} \in E \setminus \{\{k-1, k\}\}$, at least one among $\phi(l)$ and $\phi(l')$ must be strictly bigger than k and therefore $\{\phi(l), \phi(l')\}$ cannot be an element of A . The proof is now completed by applying Lemma 8.5. $\square$

With the help of Lemma 8.3 and Lemma 8.6, we are now ready to prove the following theorem, which gives an estimate for the complexity of the gradient of f .

Theorem 8.3 *Let $m := |E|$, as before. For the function f defined in Eq. (8.2.2), one can produce sets $\mathcal{D}(\epsilon)$ satisfying the criterion (8.1.1) (with $N = n(n-1)/2$) such that*

$$|\mathcal{D}(\epsilon)| \leq \exp\left(\frac{C_1 m^4 k^4 n}{\epsilon^4} \log \frac{C_2 m^4 k^4}{\epsilon^4}\right),$$

where C_1 and C_2 are universal constants.

Proof Take any $\epsilon > 0$ and let

$$\tau = \frac{\epsilon^2}{64m^2k^2}.$$

Let $\mathcal{W}(\tau)$ be as in Lemma 8.3. For each $W \in \mathcal{W}(\tau)$, let $y(W) \in \mathcal{P}_n$ be a vector such that $\|M(y) - W\| \leq n\tau$. If for some W there does not exist any such y , leave $y(W)$ undefined. Let $g_{ij} = \partial f / \partial x_{ij}$, as in Lemma 8.6. Let $g : \mathcal{P}_n \rightarrow \mathbb{R}^{n(n-1)/2}$ be the function whose (i, j) th coordinate is g_{ij} . Define

$$\mathcal{D}(\epsilon) := \{g(y) : y = y(W) \text{ for some } W \in \mathcal{W}(\tau)\}.$$

Then by Lemma 8.3

$$|\mathcal{D}(\epsilon)| \leq |\mathcal{W}(\tau)| \leq e^{34(n/\tau^2) \log(51/\tau^2)}.$$

We claim that the set $\mathcal{D}(\epsilon)$ satisfies the requirements of Theorem 8.5. To see this, take any $x \in \mathcal{P}_n$. By Lemma 8.3, there exists $W \in \mathcal{W}(\tau)$ such that $\|M(x) - W\| \leq n\tau$. In particular, this means that $y := y(W)$ is defined, and so

$$\begin{aligned} \|x - y\|_{\text{op}} &= \|M(x) - M(y)\| \\ &\leq \|M(x) - W\| + \|W - M(y)\| \\ &\leq 2n\tau. \end{aligned}$$

Therefore by Lemma 8.6,

$$\sum_{1 \leq i < j \leq n} (g_{ij}(x) - g_{ij}(y))^2 \leq 16m^2 k^2 n^2 \tau.$$

Let $z = g(x)$ and $v = g(y)$. Then $v \in \mathcal{D}(\epsilon)$, and by the above inequality,

$$\sum_{1 \leq i < j \leq n} (z_{ij} - v_{ij})^2 \leq 16m^2 k^2 n^2 \tau = \frac{n^2 \epsilon^2}{4} \leq \binom{n}{2} \epsilon^2.$$

This proves the claim that $\mathcal{D}(\epsilon)$ satisfies the requirements of Theorem 8.5. This completes the proof of Theorem 8.3. $\square$

8.3 Quantitative Estimates for Exponential Random Graphs

Take a positive integer k . Let $H_1, \dots, H_k$ be finite simple graphs and let $\beta_1, \dots, \beta_k$ be real numbers. Consider the exponential random graph on n vertices defined by the statistic T from Eq. (7.4.1) of Chap. 7. Let ψ_n be the normalizing constant of this model, as defined in Eq. (7.1.1) of Chap. 7. Let $\mathcal{P}_n$ be as in the previous section. Recall the function I defined in Eq. (8.1.2). For $x \in \mathcal{P}_n$, let

$$I(x) := \sum_{1 \leq i < j \leq n} I(x_{ij}).$$

Let

$$L_n := \sup_{x \in \mathcal{P}_n} \left(\beta_1 t(H_1, x) + \dots + \beta_k t(H_k, x) - \frac{I(x)}{n^2} \right).$$

The following theorem gives a quantitative version of Theorem 7.1 of Chap. 7 for this kind of T . This allows the possibility of extending the analysis of exponential random graphs to the sparse regime. The proof is based on an application of Theorem 8.1 using the estimates from Theorem 8.2 and Theorem 8.3.

Theorem 8.4 *Let ψ_n and L_n be as above. Let $B := 1 + |\beta_1| + \dots + |\beta_k|$. Then*

$$-\frac{cB}{n} \leq \psi_n - L_n \leq \frac{CB^{8/5}(\log n)^{1/5}}{n^{1/5}} \left(1 + \frac{\log B}{\log n} \right) + \frac{CB^2}{n^{1/2}},$$

where C and c are constants that depend only on $H_1, \dots, H_k$.

Proof Throughout this proof, C will denote any constant that depends only on the graphs $H_1, \dots, H_k$. As in the previous section, $N = n(n-1)/2$. For $1 \leq j \leq k$, define

$$T_j(x) := n^2 t(H_j, x),$$

where $t(H_j, x)$ is defined as in Eq. (8.2.1). Let

$$f(x) := \beta_1 T_1(x) + \dots + \beta_k T_k(x),$$

so that $\psi_n = F$ for this f , in the notation of Theorem 8.1. Let a , $b_{(ij)}$ and $c_{(ij)(i'j')}$ be as in Theorem 8.1, with obvious notational modification. Then

$$a \leq n^2 \sum_{r=1}^k |\beta_r| \leq CBn^2.$$

By Theorem 8.2, we get the estimates

$$b_{(ij)} \leq CB$$

and

$$c_{(ij)(i'j')} \leq \begin{cases} CBn^{-1} & \text{if } |\{i, j, i', j'\}| = 2 \text{ or } 3, \\ CBn^{-2} & \text{if } |\{i, j, i', j'\}| = 4. \end{cases}$$

Let $\mathcal{D}_1(\epsilon), \dots, \mathcal{D}_k(\epsilon)$ be the $\mathcal{D}(\epsilon)$'s for $T_1, \dots, T_k$. Define

$$\mathcal{D}(\epsilon) := \{\beta_1 d_1 + \dots + \beta_k d_k : d_r \in \mathcal{D}_r(\epsilon/\beta_r k), r = 1, \dots, k\}.$$

Clearly, for any $x \in \mathcal{P}_n$, there exists $d_1 \in \mathcal{D}_1(\epsilon/\beta_1 k), \dots, d_k \in \mathcal{D}_k(\epsilon/\beta_k k)$ such that (again, with obvious notation)

$$\begin{aligned} & \sum_{(ij)} (f_{(ij)}(x) - (\beta_1 d_{1(ij)} + \dots + \beta_k d_{k(ij)}))^2 \\ & \leq k \sum_{r=1}^k \sum_{(ij)} \beta_r^2 (T_{r(ij)}(x) - d_{r(ij)})^2 \leq N\epsilon^2. \end{aligned}$$

Therefore, $\mathcal{D}(\epsilon)$ satisfies the requirement of Theorem 8.1. Also,

$$|\mathcal{D}(\epsilon)| \leq \prod_{r=1}^k |\mathcal{D}_r(\epsilon/\beta_r k)|. \quad (8.3.1)$$

By the bounds on a , $b_{(ij)}$ and $c_{(ij)(i'j')}$ obtained in the previous section, the following estimates are easy:

$$\begin{aligned} \sum_{(ij)} ac_{(ij)(ij)} &\leq CB^2 n^3, \quad \sum_{(ij)} b_{(ij)}^2 \leq CB^2 n^2, \\ \sum_{(ij), (i'j')} ac_{(ij)(i'j')}^2 &\leq CB^3 n^3, \\ \sum_{(ij), (i'j')} b_{(ij)}(b_{(i'j')} + 4)c_{(ij)(i'j')} &\leq CB^3 n^2, \\ \sum_{(ij)} c_{(ij)(ij)}^2 &\leq CB^2, \quad \sum_{(ij)} c_{(ij)(ij)} \leq CBn. \end{aligned}$$

Combining these estimates, we see that the smoothness term is bounded by $CB^2 n^{3/2}$. Next, by (8.3.1) and Theorem 8.3,

$$\begin{aligned} \log |\mathcal{D}(\epsilon)| &\leq \sum_{r=1}^k \log |\mathcal{D}_r(\epsilon/\beta_r k)| \\ &\leq \frac{CB^4 n}{\epsilon^4} \log \frac{CB^4}{\epsilon^4}. \end{aligned}$$

Therefore, the complexity term (of Theorem 8.1) is bounded by

$$CBn^2 \epsilon + \frac{CB^4 n}{\epsilon^4} \log \frac{CB^4}{\epsilon^4}.$$

Taking

$$\epsilon = \left(\frac{B^3 \log n}{n} \right)^{1/5},$$

this gives the bound

$$CB^{8/5} n^{9/5} (\log n)^{1/5} \left(1 + \frac{\log B}{\log n} \right).$$

By Theorem 8.1, this completes the proof of the upper bound. The lower bound follows easily from Theorem 8.1 and the bound on $\sum c_{(ij)(ij)}$ obtained above. This finishes the proof of Theorem 8.4. $\square$

8.4 Nonlinear Large Deviations

Take a twice continuously differentiable $f : [0, 1]^N \rightarrow \mathbb{R}$, and let a, b_i, c_{ij} and $\mathcal{D}(\epsilon)$ be as in Sect. 8.1. Take any $p \in (0, 1)$ and let $Y = (Y_1, \dots, Y_N)$ be a vector of i.i.d. Bernoulli(p) random variables. For $x \in [0, 1]$ let

$$I_p(x) := x \log \frac{x}{p} + (1-x) \log \frac{1-x}{1-p}, \quad (8.4.1)$$

and for each $x = (x_1, \dots, x_N) \in [0, 1]^N$, let

$$I_p(x) := \sum_{i=1}^N I_p(x_i). \quad (8.4.2)$$

For $t \in \mathbb{R}$, define

$$\phi_p(t) := \inf\{I_p(x) : x \in [0, 1]^N \text{ such that } f(x) \geq tN\}. \quad (8.4.3)$$

The main result of this section, stated below, gives a sufficient condition under which the following upper tail approximation holds:

$$\mathbb{P}(f(Y) \geq tN) = \exp(-\phi_p(t) + \text{lower order terms}). \quad (8.4.4)$$

Like the approximation (8.1.3), this is also a mean field approximation. The low complexity condition on the gradient of f comes into play, as before.

Theorem 8.5 *For f as above, $p \in (0, 1)$ and Y a vector of N i.i.d. Bernoulli(p) random variables, let ϕ_p be defined as in (8.4.3). Then, for any $\delta > 0$, $\epsilon > 0$ and $t \in \mathbb{R}$,*

$$\log \mathbb{P}(f(Y) \geq tN) \leq -\phi_p(t - \delta) + \text{complexity} + \text{smoothness},$$

where, with $a, b, c_{ij}, \mathcal{D}(\epsilon)$ defined as in Sect. 8.1, and $K := \phi_p(t)/N$,

$$\begin{aligned} \text{complexity} &:= \frac{1}{4} \left(N \sum_{i=1}^N \beta_i^2 \right)^{1/2} \epsilon + 3N\epsilon + \log \left(\frac{4K \left(\frac{1}{N} \sum_{i=1}^N b_i^2 \right)^{1/2}}{\delta \epsilon} \right) \\ &\quad + \log |\mathcal{D}((\delta \epsilon)/(4K))|, \text{ and} \\ \text{smoothness} &:= 4 \left(\sum_{i=1}^N (\alpha \gamma_{ii} + \beta_i^2) + \frac{1}{4} \sum_{i,j=1}^N (\alpha \gamma_{ij}^2 + \beta_i \beta_j \gamma_{ij} + 4\beta_i \gamma_{ij}) \right)^{1/2} \\ &\quad + \frac{1}{4} \left(\sum_{i=1}^N \beta_i^2 \right)^{1/2} \left(\sum_{i=1}^N \gamma_{ii}^2 \right)^{1/2} + 3 \sum_{i=1}^N \gamma_{ii} + \log 2, \end{aligned}$$

where

$$\begin{aligned}\alpha &:= NK + N|\log p| + N|\log(1-p)|, \\ \beta_i &:= \frac{2Kb_i}{\delta} + |\log p| + |\log(1-p)|, \text{ and} \\ \gamma_{ij} &:= \frac{2Kc_{ij}}{\delta} + \frac{6Kb_ib_j}{N\delta^2}.\end{aligned}$$

Moreover,

$$\log \mathbb{P}(f(Y) \geq tN) \geq -\phi_p(t + \delta_0) - \epsilon_0 N - \log 2,$$

where

$$\epsilon_0 := \frac{1}{\sqrt{N}} \left(4 + \left| \log \frac{p}{1-p} \right| \right)$$

and

$$\delta_0 := \frac{2}{N} \left(\sum_{i=1}^N (ac_{ii} + b_i^2) \right)^{1/2}.$$

The proof of this theorem is based on Theorem 8.1. The main idea is to write the probability of $f(Y) \geq tN$ approximately as a partition function of the type considered in Theorem 8.1.

Proof (Proof of the Upper Bound in Theorem 8.5) Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a function that is twice continuously differentiable, non-decreasing, and satisfies $h(x) = -1$ if $x \leq -1$ and $h(x) = 0$ if $x \geq 0$. Let $L_1 := \|h'\|$ and $L_2 := \|h''\|$. A specific choice of h is given by

$$h(x) = 10(x+1)^3 - 15(x+1)^4 + 6(x+1)^5 - 1$$

for $-1 \leq x \leq 0$, which gives $L_1 \leq 2$ and $L_2 \leq 6$. Define

$$\psi(x) := Kh((x-t)/\delta).$$

Then clearly

$$\|\psi\| \leq K, \quad \|\psi'\| \leq \frac{L_1 K}{\delta}, \quad \|\psi''\| \leq \frac{L_2 K}{\delta^2}.$$

Let

$$g(x) := N\psi(f(x)/N) + \sum_{i=1}^N (x_i \log p + (1 - x_i) \log(1 - p)).$$

The plan is to apply Theorem 8.1 to the function g instead of f . Note that $\psi(x) = 0$ if $x \geq t$. Thus,

$$\begin{aligned} \mathbb{P}(f(Y) \geq tN) &\leq \mathbb{E}(e^{N\psi(f(Y)/N)}) \\ &= \sum_{x \in \{0,1\}^N} e^{g(x)}. \end{aligned}$$

Note also that for any $x \in [0, 1]^N$ such that $f(x) \geq tN$,

$$g(x) - I(x) = N\psi(f(x)/N) - I_p(x) = -I_p(x) \leq -\phi_p(t).$$

Again, if $f(x) \leq (t - \delta)N$, then $(f(x)/N - t)/\delta \leq -1$, and so

$$g(x) - I(x) = -NK - I_p(x) \leq -NK = -\phi_p(t).$$

Finally, note that if $f(x) = (t - \delta')N$ for some $0 < \delta' < \delta$, then

$$g(x) - I(x) \leq -I_p(x) \leq -\phi_p(t - \delta') \leq -\phi(t - \delta).$$

Combining the three cases, we see that

$$\sup_x (g(x) - I(x)) \leq -\phi_p(t - \delta).$$

Let $C_p := |\log p| + |\log(1 - p)|$. Note that

$$\|g\| \leq NK + NC_p = \alpha,$$

and for any i ,

$$\left\| \frac{\partial g}{\partial x_i} \right\| \leq \frac{2Kb_i}{\delta} + C_p = \beta_i,$$

and for any i, j ,

$$\left\| \frac{\partial^2 g}{\partial x_i \partial x_j} \right\| \leq \frac{2Kc_{ij}}{\delta} + \frac{6Kb_i b_j}{N\delta^2} = \gamma_{ij}.$$

Next, fix some $\epsilon > 0$ and let $\mathcal{D}(\epsilon)$ be as in Sect. 8.1 (for the function f). Let

$$\epsilon' := \frac{\epsilon}{2\|\psi'\|}, \quad \tau := \frac{\epsilon}{2\left(\frac{1}{N} \sum_{i=1}^N b_i^2\right)^{1/2}}.$$

Let $l \in \mathbb{R}^N$ be the vector whose coordinates are all equal to $\log(p/(1-p))$ and define

$$\mathcal{D}'(\epsilon) := \{\theta d + l : d \in \mathcal{D}(\epsilon'), \theta = j\tau \text{ for some integer } 0 \leq j < \|\psi'\|/\tau\}.$$

Let $g_i := \partial g / \partial x_i$. Take any $x \in [0, 1]^N$, and choose $d \in \mathcal{D}(\epsilon)$ satisfying (8.1.1). Choose an integer j between 0 and $\|\psi'\|/\tau$ such that

$$|\psi'(f(x)/N) - j\tau| \leq \tau.$$

Let $d' := j\tau d + l$, so that $d' \in \mathcal{D}'(\epsilon)$. Then

$$\begin{aligned} \sum_{i=1}^N (g_i(x) - d'_i)^2 &= \sum_{i=1}^N (\psi'(f(x)/N) f_i(x) - j\tau d_i)^2 \\ &\leq 2(\psi'(f(x)/N) - j\tau)^2 \sum_{i=1}^N f_i(x)^2 + 2\|\psi'\|^2 \sum_{i=1}^N (f_i(x) - d_i)^2 \\ &\leq 2\tau^2 \sum_{i=1}^N b_i^2 + 2\|\psi'\|^2 N\epsilon'^2 = N\epsilon^2. \end{aligned}$$

This shows that $\mathcal{D}'(\epsilon)$ plays the role of $\mathcal{D}(\epsilon)$ for the function g . Note that

$$|\mathcal{D}'(\epsilon)| \leq \frac{\|\psi'\|}{\tau} |\mathcal{D}(\epsilon')|.$$

This gives the upper bound on the complexity term for the function g . The proof is completed by applying Theorem 8.1. $\square$

Proof (Proof of the Lower Bound in Theorem 8.5) Fix any $z \in [0, 1]^N$ such that

$$f(z) \geq (t + \delta_0)N.$$

Let $Z = (Z_1, \dots, Z_N)$ be a random vector with independent components, where $Z_i \sim \text{Bernoulli}(z_i)$. Let $\mathcal{A}$ be the set of all $x \in \{0, 1\}^N$ such that $f(x) \geq tN$. Let g be

defined as in (8.1.17), and let $\mathcal{A}'$ be the subset of $\mathcal{A}$ where $|g(x, z) - g(x, p) - I_p(z)| \leq \epsilon_0 N$. Then

$$\begin{aligned} \mathbb{P}(f(Y) \geq tN) &= \sum_{x \in \mathcal{A}} e^{g(x, p)} \\ &= \sum_{x \in \mathcal{A}} e^{g(x, p) - g(x, z) + g(x, z)} \\ &\geq \sum_{x \in \mathcal{A}'} e^{g(x, p) - g(x, z) + g(x, z)} \geq e^{-I_p(z) - \epsilon_0 N} \mathbb{P}(Z \in \mathcal{A}'). \end{aligned} \quad (8.4.5)$$

Note that

$$\mathbb{E}(g(Z, z) - g(Z, p)) = I_p(z),$$

and

$$\begin{aligned} \text{Var}(g(Z, z) - g(Z, p)) &= \sum_{i=1}^N \text{Var}(Z_i \log(z_i/p) + (1 - Z_i) \log((1 - z_i)/(1 - p))) \\ &= \sum_{i=1}^N z_i(1 - z_i) \left(\log \frac{z_i/p}{(1 - z_i)/(1 - p)} \right)^2. \end{aligned}$$

Using the inequalities $|\sqrt{x} \log x| \leq 2/e \leq 1$ and $x(1 - x) \leq 1/4$, we see that for any $x \in [0, 1]$,

$$\begin{aligned} x(1 - x) \left(\log \frac{x/p}{(1 - x)/(1 - p)} \right)^2 &\leq \left(|\sqrt{x} \log x| + |\sqrt{1 - x} \log(1 - x)| + \frac{1}{2} \left| \log \frac{p}{1 - p} \right| \right)^2 \\ &\leq \left(2 + \frac{1}{2} \left| \log \frac{p}{1 - p} \right| \right)^2. \end{aligned}$$

Combining the last three displays, we see that

$$\begin{aligned} \mathbb{P}(|g(Z, z) - g(Z, p) - I_p(z)| > \epsilon_0 N) &\leq \frac{1}{\epsilon_0^2 N} \left(2 + \frac{1}{2} \left| \log \frac{p}{1 - p} \right| \right)^2 = \frac{1}{4}. \end{aligned} \quad (8.4.6)$$

Let $S := f(Z) - f(z)$ and $v_i(t, x) := f_i(tx + (1-t)z)$. Let $Z^{(i)}$ be defined following the same convention as in the proof of Theorem 8.1. Let $S_i := f(Z^{(i)}) - f(z)$, so that $|S - S_i| \leq b_i$. Since

$$S = \int_0^1 \sum_{i=1}^N (Z_i - z_i) v_i(t, Z) dt,$$

we have

$$\mathbb{E}(S^2) = \int_0^1 \sum_{i=1}^N \mathbb{E}((Z_i - z_i) v_i(t, Z) S) dt. \quad (8.4.7)$$

By the independence of Z_i and the pair $(S_i, Z^{(i)})$,

$$\begin{aligned} & \left| \mathbb{E}((Z_i - z_i) v_i(t, Z) S) \right| \\ &= \left| \mathbb{E}((Z_i - z_i)(v_i(t, Z) S - v_i(t, Z^{(i)}) S_i)) \right| \\ &\leq \left\| \frac{\partial v_i}{\partial x_i} \right\| \mathbb{E}|S| + \|v_i\| \mathbb{E}|S - S_i| \\ &\leq 2atc_{ii} + b_i^2. \end{aligned}$$

By (8.4.7), this gives

$$\mathbb{E}(S^2) \leq \sum_{i=1}^N (ac_{ii} + b_i^2).$$

Therefore,

$$\mathbb{P}(f(Z) < tN) \leq \frac{1}{\delta_0^2 N^2} \sum_{i=1}^N (ac_{ii} + b_i^2) = \frac{1}{4}. \quad (8.4.8)$$

Inequalities (8.4.6) and (8.4.8) give

$$\mathbb{P}(Z \in \mathcal{A}') \geq \frac{1}{2}.$$

Plugging this into (8.4.5) and taking supremum over z completes the argument. $\square$

8.5 Quantitative Estimates for Homomorphism Densities

Let $G = G(n, p)$ be an Erdős-Rényi random graph on n vertices, with edge probability p . Let H be a fixed finite simple graph. Recall the definition of the homomorphism density $t(H, G)$ from Chap. 3. Let $\mathcal{P}_n$ and $t(H, x)$ be defined as in Sect. 8.2. Recall the function I_p defined in Eq. (8.4.1), and for $x \in \mathcal{P}_n$, let

$$I_p(x) := \sum_{1 \leq i < j \leq n} I_p(x_{ij}).$$

For each $u > 1$ define

$$\psi_p(u) := \inf\{I_p(x) : x \in \mathcal{P}_n \text{ such that } t(H, x) \geq u \mathbb{E}(t(H, G))\}.$$

The following theorem shows that for any $u > 1$,

$$\mathbb{P}(t(H, G) \geq u \mathbb{E}(t(H, G))) = \exp(-\psi_p(u) + \text{lower order terms}). \quad (8.5.1)$$

provided that n is large and p is not too small. This is a quantitative version of Theorem 6.3 from Chap. 6.

Theorem 8.6 *Take any finite simple graph H and let ψ_p be defined as above. Let $X := t(H, G)$. Let k be the number of vertices and m be the number of edges of H . Suppose that $m \geq 1$ and $p \geq n^{-1/(m+3)}$. Then for any $u > 1$ and any n sufficiently large (depending only on H and u),*

$$1 - \frac{c(\log n)^{b_1}}{n^{b_2} p^{b_3}} \leq \frac{\psi_p(u)}{-\log \mathbb{P}(X \geq u \mathbb{E}(X))} \leq 1 + \frac{C(\log n)^{B_1}}{n^{B_2} p^{B_3}},$$

where c and C are constants that depend only on H and u , and

$$b_1 = 1, \quad b_2 = \frac{1}{2m}, \quad b_3 = 2m, \\ B_1 = \frac{9 + 8m}{5 + 8m}, \quad B_2 = \frac{1}{5 + 8m}, \quad B_3 = 2m - \frac{16m}{k(5 + 8m)}.$$

The remainder of this section is devoted to the proof of Theorem 8.6. The proof is a direct application of Theorem 8.5, using the estimates obtained in Sect. 8.2. The first step is to understand the properties of the rate function $\phi_p(t)$ defined in Eq. (8.4.3), put in the context of this problem. We begin with a technical lemma.

Lemma 8.7 *For any r and any $a_1, \dots, a_r, b \in [0, 1]$,*

$$\prod_{i=1}^r (a_i + b(1 - a_i)) \geq (1 - b^r) \prod_{i=1}^r a_i + b^r.$$

Proof The proof is by induction on r . The inequality is an equality for $r = 1$. Suppose that it holds for $r - 1$. Then

$$\begin{aligned}
\prod_{i=1}^r (a_i + b(1 - a_i)) &\geq \left((1 - b^{r-1}) \prod_{i=1}^{r-1} a_i + b^{r-1} \right) ((1 - b)a_r + b) \\
&= (1 - b^{r-1})(1 - b) \prod_{i=1}^r a_i + b^{r-1}(1 - b)a_r + (1 - b^{r-1})b \prod_{i=1}^{r-1} a_i + b^r \\
&\geq ((1 - b^{r-1})(1 - b) + b^{r-1}(1 - b) + (1 - b^{r-1})b) \prod_{i=1}^r a_i + b^r \\
&= (1 - b^r) \prod_{i=1}^r a_i + b^r.
\end{aligned}$$

This completes the induction. $\square$

Lemma 8.8 *Let $\phi_p(t)$ be defined as in (8.4.3), with $f(x) = n^2 t(H, x)$ and $N = n(n-1)/2$. Let l be the element of $\mathcal{P}_n$ whose coordinates are all equal to 1, and let $t_0 := f(l)/N$. Then for any $0 < \delta < t < t_0$,*

$$\phi_p(t - \delta) \geq \phi_p(t) - \left(\frac{\delta}{t_0 - t} \right)^{1/m} N \log(1/p).$$

Proof Take any $x \in \mathcal{P}_n$ such that $f(x) \geq (t - \delta)N$ and x minimizes $I_p(x)$ among all x satisfying this inequality. If $f(x) \geq tN$, then we immediately have $\phi_p(t) \leq I_p(x) = \phi_p(t - \delta)$, and there is nothing more to prove. So let us assume that $f(x) < tN$. Let

$$\epsilon := \left(\frac{tN - f(x)}{f(l) - f(x)} \right)^{1/m}.$$

For each $1 \leq i < j \leq n$, let

$$y_{ij} := x_{ij} + \epsilon(1 - x_{ij}).$$

Then $y \in \mathcal{P}_n$, and by Lemma 8.7,

$$f(y) \geq (1 - \epsilon^m)f(x) + \epsilon^m f(l) = tN.$$

Thus, by the convexity of I_p ,

$$\begin{aligned}
\phi_p(t) &\leq I_p(y) = I_p((1 - \epsilon)x + \epsilon l) \\
&\leq (1 - \epsilon)I_p(x) + \epsilon I_p(l) \\
&\leq I_p(x) + \epsilon N \log(1/p) = \phi_p(t - \delta) + \epsilon N \log(1/p).
\end{aligned}$$

Since $f(x) \geq (t - \delta)N$,

$$\epsilon^m \leq \frac{tN - (t - \delta)N}{f(l) - (t - \delta)N} \leq \frac{\delta}{t_0 - t}.$$

This completes the proof of the lemma. $\square$

Lemma 8.9 *For any p and t ,*

$$\phi_p(t) \leq \frac{1}{2}(\lceil t^{1/k} n \rceil + k)^2 \log(1/p).$$

Proof Let $r := \lceil t^{1/k} n \rceil + k$. Define $x \in [0, 1]^N$ as

$$x_{ij} := \begin{cases} 1 & \text{if } 1 \leq i < j \leq r, \\ p & \text{otherwise.} \end{cases}$$

Then

$$\begin{aligned} f(x) &\geq \frac{1}{n^{k-2}} \sum_{q \in [r]^k} \prod_{\{l, l'\} \in E} x_{qlq_{l'}} \\ &\geq \frac{r(r-1) \cdots (r-k+1)}{n^{k-2}} \geq tn^2 \geq tN, \end{aligned}$$

and since $I_p(p) = 0$,

$$I_p(x) = \sum_{i < j} I_p(x_{ij}) \leq \frac{1}{2} r^2 \log(1/p).$$

This proves the claim. $\square$

The final ingredient we need before starting the proof of Theorem 8.6 is the following concentration inequality for homomorphism densities.

Lemma 8.10 *Let X be as in the statement of Theorem 8.6 and suppose that $n > k$. Then for any $u > 1$,*

$$\mathbb{P}(X \geq u\mathbb{E}(X)) \leq e^{-Cn^2 p^{2m}},$$

where C is a positive constant that depends only on u and k .

Proof Recall that X is a function of $n(n-1)/2$ i.i.d. Bernoulli(p) random variables. If the value of one of these variables changes, then from the definition of $t(H, G)$ we see that X can change by at most n^{-2} . Moreover, observe that

$$(u-1)\mathbb{E}(X) \geq (u-1)n(n-1) \cdots (n-k+1)n^{-k}p^m,$$

since any map that takes the vertices of H to distinct vertices of G has a probability p^m of being a homomorphism. The claim now follows easily by McDiarmid's inequality (Theorem 4.3). $\square$

We are now ready to prove Theorem 8.6.

Proof (Proof of the Upper Bound in Theorem 8.6) The task now is to pull together all the information obtained above, for use in Theorem 8.5. As before, we work with $f(x) = n^2 t(H, x)$. Take $t = \kappa p^m$ for some fixed $\kappa > 0$. Let δ and ϵ be two positive real numbers, both less than t , to be chosen later. Note that $\delta < t < \kappa p^{2m/k}$ since $t = \kappa p^m$ and $k > 2$. Assume that δ and ϵ are bigger than $n^{-1/2}$. Note that p is already assumed to be bigger than $n^{-1/2}$ in the statement of the theorem.

Recall that the indexing set for quantities like b_i and c_{ij} , instead of being $\{1, \dots, N\}$, is now $\{(i, j) : 1 \leq i < j \leq n\}$. For simplicity, we will write (ij) instead of (i, j) . Throughout, C will denote any constant that depends only on the graph H , the constant κ , and nothing else. From Theorem 8.2, we have the estimates

$$a \leq n^2, \quad b_{(ij)} \leq C,$$

and

$$c_{(ij)(i'j')} \leq \begin{cases} Cn^{-1} & \text{if } |\{i, j, i', j'\}| = 2 \text{ or } 3, \\ Cn^{-2} & \text{if } |\{i, j, i', j'\}| = 4. \end{cases}$$

Let $\theta := \delta^{-1} p^{2m/k}$. By Lemma 8.9,

$$K \leq Cp^{2m/k} \log n.$$

Using the above bounds, we get

$$\alpha \leq Cn^2 \log n, \quad \beta_{(ij)} \leq C\theta \log n,$$

and

$$\gamma_{(ij)(i'j')} \leq \begin{cases} Cn^{-1} \theta \log n & \text{if } |\{i, j, i', j'\}| = 2 \text{ or } 3, \\ Cn^{-2} \delta^{-1} \theta \log n & \text{if } |\{i, j, i', j'\}| = 4. \end{cases}$$

Therefore, we have the estimates

$$\sum_{(ij)} \beta_{(ij)}^2 \leq Cn^2 \theta^2 (\log n)^2, \quad \sum_{(ij)} b_{(ij)}^2 \leq Cn^2,$$

and by Theorem 8.3,

$$\begin{aligned} \log |\mathcal{D}((\delta\epsilon)/(4K))| &\leq \frac{Cn\theta^4}{\epsilon^4} \log \frac{CK}{\delta\epsilon} \\ &\leq \frac{Cn\theta^4(\log n)^5}{\epsilon^4}. \end{aligned}$$

Combining the last three estimates, we see that the complexity term in Theorem 8.5 is bounded above by

$$Cn^2\epsilon\theta \log n + \frac{Cn\theta^4(\log n)^5}{\epsilon^4}.$$

Taking $\epsilon = n^{-1/5}\theta^{3/5}(\log n)^{4/5}$, the above bound simplifies to

$$Cn^{9/5}\theta^{8/5}(\log n)^{9/5}.$$

Next, note that by the bounds obtained above and the inequality $\delta > n^{-1/2}$,

$$\begin{aligned} \sum_{(ij)} \alpha \gamma_{(ij)(ij)} &\leq Cn^3\theta(\log n)^2, \\ \sum_{(ij),(i'j')} \alpha \gamma_{(ij)(i'j')}^2 &\leq Cn^3\theta^2(\log n)^3, \\ \sum_{(ij),(i'j')} \beta_{(ij)}(\beta_{(i'j')} + 4)\gamma_{(ij)(i'j')} &\leq Cn^2\delta^{-1}\theta^3(\log n)^3, \\ \left(\sum_{(ij)} \beta_{(ij)}^2 \right)^{1/2} \left(\sum_{(ij)} \gamma_{(ij)(ij)}^2 \right)^{1/2} &\leq Cn\theta^2(\log n)^2, \\ \sum_{(ij)} \gamma_{(ij)(ij)} &\leq Cn\theta \log n. \end{aligned}$$

The above estimates show that the smoothness term in Theorem 8.5 is bounded above by a constant times

$$n^{3/2}\theta(\log n)^{3/2} + n\delta^{-1/2}\theta^{3/2}(\log n)^{3/2} + n\theta^2(\log n)^2.$$

Putting $\eta := p^{2m/k}$, we see that this is bounded by a constant times

$$n^{3/2}\delta^{-1}\eta(\log n)^{3/2} + n\delta^{-2}\eta^{3/2}(\log n)^2.$$

Since $\delta > n^{-1/2}$, we can further simplify this upper bound to

$$n^{3/2}\delta^{-1}\eta(\log n)^2.$$

Combining the bounds on the complexity term and the smoothness term, we get that

$$\begin{aligned} \log \mathbb{P}(f(Y) \geq tN) &\leq -\phi_p(t - \delta) + Cn^{9/5}\delta^{-8/5}\eta^{8/5}(\log n)^{9/5} \\ &\quad + Cn^{3/2}\delta^{-1}\eta(\log n)^2. \end{aligned}$$

By Lemma 8.8,

$$-\phi_p(t - \delta) \leq -\phi_p(t) + C\delta^{1/m}n^2 \log n.$$

Taking

$$\delta = n^{-m/(5+8m)}\eta^{8m/(5+8m)}(\log n)^{4m/(5+8m)}$$

gives

$$\begin{aligned} \log \mathbb{P}(f(Y) \geq tN) & \tag{8.5.2} \\ &\leq -\phi_p(t) + Cn^{(9+16m)/(5+8m)}\eta^{8/(5+8m)}(\log n)^{(9+8m)/(5+8m)} \\ &\quad + Cn^{(15+26m)/(10+16m)}\eta^{5/(5+8m)}(\log n)^{(10+12m)/(5+8m)}. \end{aligned}$$

Now note that since $p > n^{-1/2}$,

$$\begin{aligned} \frac{n^{(9+16m)/(5+8m)}\eta^{8/(5+8m)}}{n^{(15+26m)/(10+16m)}\eta^{5/(5+8m)}} &= n^{(3+6m)/(10+16m)}p^{6m/(5k+8mk)} \\ &\geq n^{(3+6m)/(10+16m)}n^{-3m/(5+8m)} \\ &= n^{3/(10+16m)}. \end{aligned}$$

This shows that the first term on the right-hand side in (8.5.2) dominates the second when n is sufficiently large. Therefore, when n is large enough,

$$\begin{aligned} \log \mathbb{P}(f(Y) \geq tN) & \\ &\leq -\phi_p(t) + Cn^{(9+16m)/(5+8m)}p^{16m/(5k+8mk)}(\log n)^{(9+8m)/(5+8m)}. \end{aligned}$$

Written differently, this is

$$\begin{aligned} & \frac{\phi_p(t)}{-\log \mathbb{P}(f(Y) \geq tN)} \\ & \leq 1 + \frac{Cn^{(9+16m)/(5+8m)}p^{16m/(5k+8mk)}(\log n)^{(9+8m)/(5+8m)}}{-\log \mathbb{P}(f(Y) \geq tN)}. \end{aligned}$$

By Lemma 8.10,

$$-\log \mathbb{P}(f(Y) \geq tN) \geq Cn^2p^{2m}, \quad (8.5.3)$$

Therefore,

$$\begin{aligned} & \frac{\phi_p(t)}{-\log \mathbb{P}(f(Y) \geq tN)} \\ & \leq 1 + Cn^{-1/(5+8m)}p^{-2m+16m/(5k+8mk)}(\log n)^{(9+8m)/(5+8m)}. \end{aligned}$$

A minor verification using the assumption $p \geq n^{-1/4}$ shows that the ϵ and δ chosen above are both bigger than $n^{-1/2}$, as required. To complete the proof of the upper bound, notice that $\mathbb{E}(X)$ is asymptotic to p^m since $p \geq n^{-1/(m+3)}$. $\square$

Proof (Proof of the Lower Bound in Theorem 8.6) First, observe that by Lemma 8.8, Theorem 8.2, and the lower bound in Theorem 8.5,

$$\log \mathbb{P}(f(Y) \geq tN) \geq -\phi_p(t) - Cn^{-1/2m}n^2 \log n.$$

Therefore, again applying (8.5.3), we get

$$\frac{\phi_p(t)}{-\log \mathbb{P}(f(Y) \geq tN)} \geq 1 - Cn^{-1/2m}p^{-2m} \log n.$$

This completes the proof of the lower bound. $\square$

8.6 Explicit Rate Function for Triangles

Theorem 8.6 converts the large deviation question for subgraph densities into a question of solving a variational problem analogous to the variational problem for dense graphs that we saw in Theorem 5.2. Unlike the dense case, however, the sparse regime allows us to explicitly solve the variational problem in many examples. The simplest example, that of triangle density, is solved in this section. We will prove the following result.

Theorem 8.7 *Let $T_{n,p}$ be the number of triangles in an Erdős–Rényi graph $G(n, p)$. Then for any fixed $\delta > 0$, as $n \rightarrow \infty$ and $p \rightarrow 0$ simultaneously such that $p \gg n^{-1/158}(\log n)^{33/158}$,*

$$\begin{aligned} \mathbb{P}(T_{n,p} \geq (1 + \delta)\mathbb{E}(T_{n,p})) \\ = \exp\left(- (1 - o(1)) \min\left\{\frac{\delta^{2/3}}{2}, \frac{\delta}{3}\right\} n^2 p^2 \log(1/p)\right). \end{aligned}$$

Let H be a triangle, and for simplicity, let $T(f) := t(H, f)$. Let $\mathcal{P}_n$ and $t(H, x)$ be as in Sect. 8.2. For $x \in \mathcal{P}_n$, let $T(x) := t(H, x)$. For $\delta > 0$, let

$$\psi_p(n, \delta) := \inf\{I_p(x) : x \in \mathcal{P}_n, T(x) \geq (1 + \delta)p^3\}.$$

By Theorem 8.6, it suffices to show that under the given conditions,

$$\frac{\psi_p(n, \delta)}{n^2 p^2 \log(1/p)} \rightarrow \min\left\{\frac{\delta^{2/3}}{2}, \frac{\delta}{3}\right\}. \quad (8.6.1)$$

The remainder of this section is devoted to the proof of (8.6.1). The proof requires a number of technical estimates, which are worked out below. Throughout, we will use the notation $a \ll b$ to mean that the two quantities a and b are varying in such a way that $a/b \rightarrow 0$. Similarly, $a \sim b$ will mean that $a/b \rightarrow 1$. The symbol $o(a)$ will denote any quantity that, when divided by a , tends to zero as the relevant parameters vary in some prescribed manner. Let I_p be defined as in (8.4.1).

Lemma 8.11 *If $p \rightarrow 0$ and $0 \leq x \ll p$, then $I_p(p + x) \sim x^2/(2p)$. On the other hand, if $p \rightarrow 0$, $x \rightarrow 0$ and $p \ll x$, then $I_p(p + x) \sim x \log(x/p)$.*

Proof Note that $I_p(p) = I'_p(p) = 0$, $I''_p(p) = 1/(p(1 - p))$ and

$$I'''_p(x) = \frac{1}{(1 - x)^2} - \frac{1}{x^2}. \quad (8.6.2)$$

The first assertion of the lemma follows easily by Taylor expansion, using the above identities. For the second assertion, note that if $p \rightarrow 0$, $x \rightarrow 0$ and $p \ll x$, then

$$\begin{aligned} I_p(p + x) &= (p + x) \log \frac{p + x}{p} + (1 - p - x) \log \frac{1 - p - x}{1 - p} \\ &= x \log \frac{x}{p} + O(x). \end{aligned}$$

This completes the proof of the lemma. $\square$

Lemma 8.12 *There exists $p_0 > 0$ such that for all $0 < p \leq p_0$ and $0 \leq x \leq b \leq 1 - p - 1/\log(1/p)$,*

$$I_p(p + x) \geq (x/b)^2 I_p(p + b).$$

Proof Let $q := 1 - p - 1/\log(1/p)$. Let $f(t) := I_p(p + \sqrt{t})$. Note that

$$\begin{aligned} f''(t) &= \frac{d}{dt} \left(\frac{1}{2\sqrt{t}} I'_p(p + \sqrt{t}) \right) \\ &= \frac{1}{4t} I''_p(p + \sqrt{t}) - \frac{1}{4t^{3/2}} I'_p(p + \sqrt{t}). \end{aligned}$$

Let $g(x) := 4x^3 f''(x^2)$. Then by the above formula,

$$g(x) = x I''_p(p + x) - I'_p(p + x).$$

By (8.6.2), this shows that

$$g'(x) = x I'''_p(p + x) \quad \begin{cases} \leq 0 & \text{if } p + x \leq 1/2, \\ \geq 0 & \text{if } p + x \geq 1/2. \end{cases} \quad (8.6.3)$$

Now note that $g(0) = 0$, and as $p \rightarrow 0$,

$$\begin{aligned} g(q) &= \frac{q \log(1/p)}{(1 - 1/\log(1/p))} - \log \frac{(1 - p)(\log(1/p) - 1)}{p} \\ &= -\log \log(1/p) + O(1). \end{aligned}$$

Thus, there exists p_0 small enough such that if $p \leq p_0$, then $g(q) < 0$. By (8.6.3), this implies that $g(x) \leq 0$ for all $x \in [0, q]$. In particular, f is concave in the interval $[0, q^2]$. Thus, for any $0 \leq x \leq b \leq q$,

$$I_p(p + x) = f(x^2) \geq (x^2/b^2) f(b^2) = (x/b)^2 I_p(p + b),$$

which completes the proof. $\square$

Corollary 8.1 *There is some $p_0 > 0$ such that when $0 < p \leq p_0$ and $0 \leq x \leq 1 - p$,*

$$I_p(p + x) \geq x^2 I_p(1 - 1/\log(1/p)).$$

Proof Let p_0 be as in Lemma 8.12 and let $b = 1 - p - 1/\log(1/p)$. When $0 \leq x \leq b$, the inequality follows by Lemma 8.12 since $b < 1$. When $b \leq x \leq 1 - p$, then we trivially deduce $I_p(p + x) \geq I_p(p + b) \geq x^2 I_p(p + b)$. $\square$

Now let

$$\phi_p(\delta) := \frac{1}{2} \inf\{I_p(f) : f \in \mathcal{W}, T(f) \geq (1 + \delta)p^3\}.$$

Note that unlike $\psi_p(n, \delta)$, $\phi_p(\delta)$ does not depend on n . The following lemma compares $\psi_p(n, \delta)$ and $\phi_p(\delta)$.

Lemma 8.13 *For any p , n and δ ,*

$$\phi_p(\delta) \leq \frac{1}{n^2} \psi_p(n, \delta) + \frac{1}{2n} \log \frac{1}{1-p}.$$

Proof For any $x \in \mathcal{P}_n$, construct a graphon f in the natural way, by defining $f(s, t) = x_{ij}$ if either (s, t) or (t, s) belongs to the square $[(i-1)/n, i/n) \times [(j-1)/n, j/n)$ for some $1 \leq i \leq j \leq n$, where x_{ii} is taken to be zero. It is easy to verify that $T(f) = T(x)$, and

$$n^2 I_p(f) = 2I_p(x) + nI_p(0).$$

Dividing by $2n^2$ on both sides and taking infimum over x completes the proof. $\square$
We are now ready to prove Theorem 8.7.

Proof (Proof of Theorem 8.7) Throughout this proof, C will denote any universal constant and $o(1)$ will denote any quantity that depends only on p and tends to zero as $p \rightarrow 0$.

Take any f such that $T(f) \geq (1 + \delta)p^3$. First, we wish to show that

$$\frac{1}{2} I_p(f) \geq (1 - o(1)) \min \left\{ \frac{\delta^{2/3}}{2}, \frac{\delta}{3} \right\} p^2 \log(1/p). \quad (8.6.4)$$

By Lemma 8.13, this will show that

$$\liminf \frac{\psi_p(n, \delta)}{n^2 p^2 \log(1/p)} \geq \min \left\{ \frac{\delta^{2/3}}{2}, \frac{\delta}{3} \right\},$$

proving one direction of (8.6.1). Assume that

$$\frac{1}{2} I_p(f) \leq \min \left\{ \frac{\delta^{2/3}}{2}, \frac{\delta}{3} \right\} p^2 \log(1/p). \quad (8.6.5)$$

We can make this assumption because if there exists no such f (among all f satisfying $T(f) \geq (1 + \delta)p^3$), then the proof of (8.6.4) is automatically complete.

Since I_p is decreasing in $[0, p]$ and increasing in $[p, 1]$, we may assume without loss of generality that $f \geq p$ everywhere. (Otherwise, it suffices to prove (8.6.4)

for $f' := \max\{f, p\}$, because $T(f') \geq T(f) \geq (1 + \delta)p^3$ and $I_p(f') \leq I_p(f)$.) Let $g := f - p$. Then

$$T(f) = T(p + g) = p^3 + T(g) + 3pR(g) + 3p^2S(g), \quad (8.6.6)$$

where

$$R(g) := \int_{[0,1]^3} g(x, y)g(y, z) dx dy dz$$

and

$$S(g) := \int_{[0,1]^2} g(x, y) dx dy.$$

Now, if $S(g) \geq p^{3/2} \log(1/p)$, then by Lemma 8.11,

$$\begin{aligned} I_p(f) &= I_p(p + g) \geq I_p(p + S(g)) \\ &\geq I_p(p + p^{3/2} \log(1/p)) \gg p^2 \log(1/p), \end{aligned}$$

which proves (8.6.4). So let us assume that

$$S(g) < p^{3/2} \log(1/p). \quad (8.6.7)$$

Under the above assumption, (8.6.6) gives

$$T(g) + 3pR(g) \geq (\delta - o(1))p^3. \quad (8.6.8)$$

For each $x \in [0, 1]$, let

$$d(x) := \int_0^1 g(x, y) dy,$$

and for each $b \in [0, 1]$, let

$$B_b := \{x \in [0, 1] : d(x) \geq b\}.$$

Note that by Jensen's inequality,

$$I_p(f) = I_p(p + g) \geq \int_0^1 I_p(p + d(x)) dx.$$

Thus,

$$\text{Leb}(B_b) \leq \frac{\int_0^1 I_p(p + d(x)) dx}{I_p(p + b)} \leq \frac{I_p(f)}{I_p(p + b)}.$$

Therefore by the assumption (8.6.5), we get

$$\text{Leb}(B_b) \leq \frac{Cp^2 \log(1/p)}{I_p(p+b)}.$$

If b varies with p such that $p \ll b < 1-p$, then by Lemma 8.11, the above inequality gives

$$\text{Leb}(B_b) \leq \frac{Cp^2 \log(1/p)}{b \log(b/p)} \leq \frac{Cp^2}{b}. \quad (8.6.9)$$

Thus,

$$\int_{B_b \times B_b} g(x, y)^2 dx dy \leq \text{Leb}(B_b)^2 \leq \frac{Cp^4}{b^2}. \quad (8.6.10)$$

Let D be the set of all $(x, y, z) \in [0, 1]^3$ such that at least one of x, y and z belongs to B_b . Choose b depending on p such that $\sqrt{p \log(1/p)} \ll b \ll 1$. Then by (8.6.7) and (8.6.9),

$$\int_D g(x, y)g(y, z)g(z, x) dx dy dz \leq 3\text{Leb}(B_b)S(g) \ll p^3. \quad (8.6.11)$$

Let $B_b^c := [0, 1] \setminus B_b$, and let

$$\delta_1 := \frac{1}{p^2} \int_{B_b \times B_b^c} g(x, y)^2 dx dy$$

and

$$\delta_2 := \frac{1}{p^2} \int_{B_b^c \times B_b^c} g(x, y)^2 dx dy.$$

By the inequality (8.6.11) and the generalized Hölder inequality from Chap. 2 (Theorem 2.1),

$$\begin{aligned} T(g) &= \int_{B_b^c \times B_b^c \times B_b^c} g(x, y)g(y, z)g(z, x) dx dy dz + o(p^3) \\ &\leq (\delta_2^{3/2} + o(1))p^3. \end{aligned} \quad (8.6.12)$$

Next, by Lemma 8.12 and Jensen's inequality,

$$\begin{aligned} I_p(f) &= I_p(p + g) \geq \int_0^1 I_p(p + d(x)) dx \\ &\geq \int_{B_b^c} I_p(p + d(x)) dx \\ &\geq \frac{I_p(p + b)}{b^2} \int_{B_b^c} d(x)^2 dx. \end{aligned}$$

Proceeding as in the proof of (8.6.9), this gives

$$\int_{B_b^c} d(x)^2 dx \leq \frac{b^2 I_p(f)}{I_p(p + b)} \leq Cp^2 b = o(p^2).$$

Therefore,

$$\begin{aligned} R(g) &= \int_0^1 d(x)^2 dx = \int_{B_b} d(x)^2 dx + o(p^2) \\ &\leq \int_{B_b} \int_0^1 g(x, y)^2 dy dx + o(p^2). \end{aligned}$$

Combining this with Eq. (8.6.10), we get

$$R(g) \leq (\delta_1 + o(1))p^2. \quad (8.6.13)$$

By Eqs. (8.6.8), (8.6.12) and (8.6.13), we get

$$3\delta_1 + \delta_2^{3/2} \geq \delta - o(1).$$

Finally, by Corollary 8.1,

$$I_p(f) = I_p(p + g) \geq (1 - o(1))(2\delta_1 + \delta_2)p^2 \log(1/p),$$

where the $o(1)$ term appears because

$$I_p(1 - 1/\log(1/p)) = (1 - o(1)) \log(1/p).$$

Combining all of the above, we get

$$\begin{aligned} \liminf_{p \rightarrow 0} \frac{\phi_p(\delta)}{p^2 \log(1/p)} &\geq \inf \left\{ \delta_1 + \frac{1}{2}\delta_2 : \delta_1 \geq 0, \delta_2 \geq 0, \right. \\ &\quad \left. 3\delta_1 + \delta_2^{3/2} \geq \delta \right\}. \end{aligned} \quad (8.6.14)$$

Now take any $a \geq 0$, and for $0 \leq x \leq a/3$, let

$$f(x) := x + \frac{1}{2}(a - 3x)^{2/3}.$$

An easy verification shows that depending on the value of a , either f is increasing throughout the interval $[0, a/3]$, or increases up to a maximum value and then decreases. In either case, the minimum value of f is achieved at one of the two endpoints of the interval $[0, a/3]$. This shows that under the constraint $3\delta_1 + \delta_2^{3/2} = a$, the minimum value of $\delta_1 + \delta_2/2$ is attained at either $\delta_1 = 0$ or $\delta_2 = 0$. Since this is true for any a , the minimization in (8.6.14) is attained when either $\delta_1 = 0$ or $\delta_2 = 0$. From this, (8.6.4) follows easily.

Let us now prove the other direction of (8.6.1), that is,

$$\limsup \frac{\psi_p(n, \delta)}{n^2 p^2 \log(1/p)} \leq \min \left\{ \frac{\delta^{2/3}}{2}, \frac{\delta}{3} \right\}. \quad (8.6.15)$$

First let a be an integer, to be chosen later, and let $x \in \mathcal{P}_n$ be the element defined as $x_{ij} = 1$ if $1 \leq i < j \leq a$, and $x_{ij} = p$ otherwise. Then it is easy to see that

$$T(x) \geq \frac{1}{n^3} (a(a-1)(a-2) + (n(n-1)(n-2) - a(a-1)(a-2))p^3)$$

and

$$I_p(x) = \frac{a(a-1)}{2} \log(1/p).$$

Thus, we can choose a such that $a = (\delta^{1/3} + o(1))np$, $T(x) \geq (1 + \delta)p^3$, and

$$I_p(x) = \left(\frac{\delta^{2/3}}{2} + o(1) \right) n^2 p^2 \log(1/p).$$

Next, let x be defined as $x_{ij} = 1$ if $1 \leq i \leq a$ and $i < j$, and $x_{ij} = p$ otherwise. Then

$$T(x) \geq \frac{1}{n^3} (3a(n-a)(n-a-1)p + (n-a)(n-a-1)(n-a-2)p^3)$$

and

$$I_p(x) = a \left(n - \frac{a+1}{2} \right) \log(1/p).$$

Thus, we can choose a such that $a = (\delta/3 + o(1))p^2n$, $T(x) \geq (1 + \delta)p^3$ and

$$I_p(x) = \left(\frac{\delta}{3} + o(1)\right)n^2p^2 \log(1/p).$$

Combining the two bounds proves (8.6.15). $\square$

Bibliographical Notes

The theory developed in the earlier chapters applies only to dense random graphs. However, most graphs that arise in applications are sparse. The main roadblock to developing the analogous theory for sparse graphs is the unavailability of a suitable sparse version of graph limit theory. Although a number of attempts have been made towards extending graph limit theory to the sparse setting such as in Bollobás and Riordan [2] and Borgs et al. [3, 4], an effective version of a sparse ‘counting lemma’ that connects the cut metric with subgraph counts, is still missing.

There are numerous powerful upper bounds for tail probabilities of nonlinear functions of independent Bernoulli random variables, such as the bounded difference inequality popularized by McDiarmid [18], and the inequalities of Talagrand [19], Latała [16], Kim and Vu [14, 15], Vu [20] and Janson et al. [13]. However, these upper bounds generally do not yield the right constants in the exponent, which is the goal of large deviations theory. Even establishing the right powers of n and p in the rate functions for subgraph densities has been quite challenging—see, for example, Chatterjee [7] and DeMarco and Kahn [10, 11].

Motivated by the problem of understanding the large deviations of subgraph densities in sparse random graphs, the nonlinear large deviation theory presented in this chapter was developed in Chatterjee and Dembo [8]. The technique used for proving Lemmas 8.1 and 8.2 was originally developed in Chatterjee [5, 6] and Chatterjee and Dey [9] in the context of developing rigorous formulations and proofs of mean field equations. The move from mean field equations to the mean field approximation of Theorem 8.1 required the low complexity gradient condition, which was the main new idea in Chatterjee and Dembo [8]. The proof of Theorem 8.6 presented here is slightly shorter than the one in [8], eliminating the need for using a result of Janson et al. [13], which would have required a large number of additional pages if I attempted to present the proof here. This comes at the cost of a slightly worse error bound.

Theorem 8.7 was proved in Lubetzky and Zhao [17]. The proof presented here is almost a verbatim copy of the proof in Lubetzky and Zhao [17]. The error bounds in Chatterjee and Dembo [8] allow p to go to zero slower than $n^{-1/42}$ in Theorem 8.7. The slightly worse error bound of Theorem 8.6 is the reason why p needs to decay slower than $n^{-1/158}$ here. However, these are all very far from the conjectured threshold: It is believed that Theorem 8.7 holds whenever $p \rightarrow 0$ slower than $n^{-1/2}$. Incidentally, Lubetzky and Zhao [17] show that with the use of a weak version of

Szemerédi's regularity lemma, it is possible to probe a sparse large deviation result as long as p tends to zero slower than a negative power of $\log n$.

Lubetzky and Zhao [17] also proved analogous results for the density of cliques. These results were extended to general homomorphism densities by Bhattacharya et al. [1], who obtained the following beautiful explicit formula for the rate function for general homomorphism densities. Take any finite simple graph H with maximum degree Δ . Let H^* be the induced subgraph of H on all vertices whose degree in H is Δ . Recall that an independent set in a graph is a set of vertices such that no two are connected by an edge. Also, recall that a graph is called regular if all its vertices have the same degree, and irregular otherwise. Define a polynomial

$$P_{H^*}(x) := \sum_k i_{H^*}(k) x^k,$$

where $i_{H^*}(k)$ is the number of k -element independent sets in H^* .

Theorem 8.8 (Bhattacharya et al. [1]) *Let H be a connected finite simple graph on k vertices with maximum degree $\Delta \geq 2$. Then for any $\delta > 0$, there is a unique positive number $\theta = \theta(H, \delta)$ that solves $P_{H^*}(\theta) = 1 + \delta$, where P_{H^*} is the polynomial defined above. Let $H_{n,p}$ be the number of homomorphisms of H into an Erdős–Rényi $G(n, p)$ random graph. Then there is a constant $\alpha_H > 0$ depending only on H , such that if $n \rightarrow \infty$ and $p \rightarrow 0$ slower than $n^{-\alpha_H}$, then for any $\delta > 0$,*

$$\mathbb{P}(H_{n,p} \geq (1 + \delta)\mathbb{E}(H_{n,p})) = \exp\left(- (1 + o(1))c(\delta)n^2 p^\Delta \log \frac{1}{p}\right),$$

where

$$c(\delta) = \begin{cases} \min\{\theta, \frac{1}{2}\delta^{2/k}\} & \text{if } H \text{ is regular,} \\ \theta & \text{if } H \text{ is irregular.} \end{cases}$$

The formula given in Theorem 8.8 is more than just a formula. It gives a hint at the conditional structure of the graph, and at the nature of phase transitions as δ varies. Unlike the dense case, it is hard to give a precise meaning to claims about the conditional structure in the sparse setting due to the lack of an adequate sparse graph limit theory.

The paper [1] also gives a number of examples where the coefficient $c(\delta)$ in Theorem 8.8 can be explicitly computed. For instance, if $H = C_4$, the cycle of length four, then

$$c(\delta) = \begin{cases} \frac{1}{2}\sqrt{\delta} & \text{if } \delta < 16, \\ -1 + \sqrt{1 + \frac{1}{2}\delta} & \text{if } \delta \geq 16. \end{cases}$$

Theorem 8.7 is also a special case of Theorem 8.8.

A very interesting recent development in nonlinear large deviations is the preprint of Eldan [12]. Eldan's proof technique is quite different than the one presented here. Eldan's approach is also based on the low complexity gradient condition, although the complexity of the gradient is measured differently, as follows. Recall that the Gaussian width of a measurable set $K \subseteq \mathbb{R}^N$ is defined as

$$\text{GW}(K) := \mathbb{E}(\sup_{x \in K} x \cdot Z),$$

where Z denotes a standard Gaussian random vector and $x \cdot Z$ denotes the inner product of x and Z . Eldan measure the complexity of the gradient of a function $f : \{0, 1\}^N \rightarrow \mathbb{R}$ as

$$\mathcal{D}(f) := \text{GW}(\{\Delta f(x) : x \in \{0, 1\}^N\} \cup \{0\}),$$

where $\Delta f(x) = (\Delta_1 f(x), \dots, \Delta_N f(x))$ is the discrete gradient of f , with $\Delta_i f(x)$ defined as in (8.1.4). Eldan measures the smoothness of f using the Lipschitz constant $\text{Lip}(f)$ of f with respect to the Hamming distance on $\{0, 1\}^N$. In terms of the discrete derivatives, $\text{Lip}(f) = \max_{i,x} |\Delta_i f(x)|$. One of the main results of Eldan's paper is the following improved version of Theorem 8.5.

Theorem 8.9 (Eldan [12]) *Let all notation be as in Theorem 8.5. Let $\text{Lip}(f)$ denote the Lipschitz constant of f with respect to the Hamming distance on $\{0, 1\}^N$. Then for any $t, \delta \in \mathbb{R}$ such that $0 < \delta < \phi_p(t - \delta)/n$,*

$$\log \mathbb{P}(f(Y) \geq tN) \leq -\phi_p(t - \delta)(1 - 6\delta^{-1}ABN^{-1/3}(\log N)^{1/6}),$$

where

$$A = \left(\text{Lip}(f) + \frac{1}{4\delta\sqrt{n}}\text{Lip}(f)^2 + |\log(p(1-p))| \right)^{2/3},$$

$$B = \left(2\mathcal{D}(f) + \frac{1}{4\delta}\text{Lip}(f)^2 \right)^{1/3}.$$

Moreover, if $\text{Lip}(f)^2 \leq n\delta^2$, then

$$\log \mathbb{P}(f(Y) \geq tN) \geq -\phi_p(t) \left(1 + \frac{1}{2n\delta^2}\text{Lip}(f)^2 \right) - \log 10.$$

The quantities A and B correspond to the smoothness and complexity terms in Theorem 8.5. Theorem 8.9 has several advantages over Theorem 8.5. First, it has a cleaner statement. Second, it does not use the second derivatives of f at all. Third, it gives improved error bounds. For example, using Theorem 8.9, Eldan [12] demonstrated that Theorem 8.7 holds whenever $p \rightarrow 0$ slower than $n^{-1/18}$. This, unfortunately, is still far from the conjectured optimal threshold of $n^{-1/2}$.

Lastly, another interesting recent development is the paper of Yan [21], which extends the nonlinear large deviation theory to functions of a more general class of random variables than Bernoulli. In particular, Yan's theorems apply to continuous random variables.

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